

Part Selection Table

Switching Frequency	265 kHz
FET Selection - High Side Switching Assumed High Side Switching Driver Available	NTMFS5844NL/ NVMFS5844NL
Diode Selection	PMEG045V150EPD
Heat Sink Solution	FET and Diode each have the same design. Assuming 12 mil via and 62 mil board. Via Farm (16 12-mil vias, 0.6mm center to center spacing) guiding heat from Diode / FET through the PCB to the other side to the thermal pad, then to the heatsink. >=50 PSI of Pressure Force against Ultrasoft Thermal Pad. (TG-A1780). BGA push pin heat sink, mounted on the other side of the PCB. (HSB28-606022) See Thermal Section for Detailed Calculation R_diode_JA = 32.31 C/W R_FET_JA = 27.254 C/W

Controller Transfer Function - PI Controller with 2 1st order LPF to suppress resonant.

Note that 20% overshoot requirements correspond to > 45 degree of phase margin in context of frequency domain control. Crossover point of the open loop transfer function must have a phase margin larger than this value.

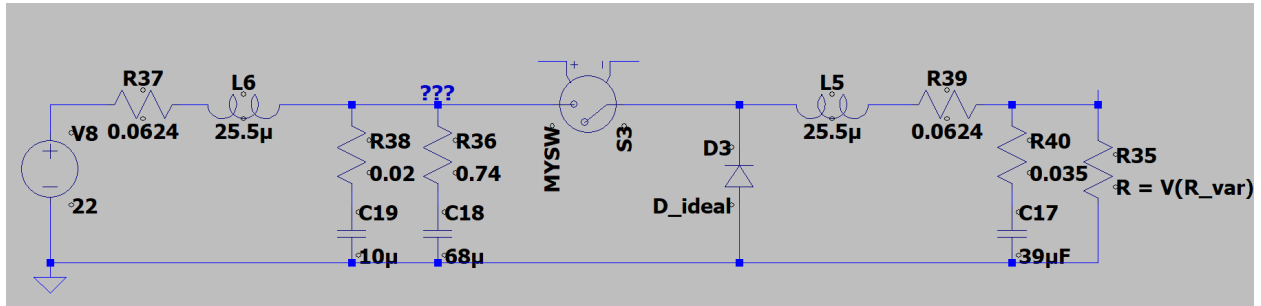
Performance Report Section include phase margin plots. See Controller & Close Loop Simulation Section for more details (and SPICE simulation)

$$Controller(s) = \frac{2200}{5200} \cdot \left(\frac{1}{4.7 \cdot 10^{-6} \cdot 680 \cdot s} \right) \cdot \left(\frac{10000}{10000} \cdot \frac{1}{10000 \cdot (6.8 \cdot 10^{-9}) \cdot s + 1} \right) \cdot \left(\frac{10000}{10000} \cdot \frac{1}{10000 \cdot (6.8 \cdot 10^{-9}) \cdot s + 1} \right)$$

Type	Designator	Value	Extra Notes
Input EMI Filters - L, C and Rd-Cd network			
Input Inductor	Lin, Lf	25.5uH (Derived)	
Lin Core Type		RM12/I3F3/A315	
Inductor Wire Gauge		10 AWG - D = 2.588 mm	mOhm/m = 3.277
Turns	N_turn_in	9	
Derived Wire Resistance DC		60.24 mOhm	
Input Filter Capacitor	Cin, Cf	10u, ESR = 20 mOhm, 50V rated.	Multilayer Ceramics KTS500B106M55N0T00
Rd-Cd Network		Rd =0.74 , Cd = 68u	Cd is Aluminum Cap. Rated for 50V. 50SVF68M ESR = 20 mOhm
Filter Characteristics Frequency		~10 kHz	Zo = 1.6 Ohm

Type	Designator	Value	Extra Notes
Output L and C			
Output Inductor	Lout	25.5uH (Derived)	*Identical Design to Input Inductor
Lout Core Type		RM12/I3F3/A315	
Inductor Wire Gauge		10 AWG - D = 2.588 mm	mOhm/m = 3.277
Turns	N_turn_out	9	R_DC= 0.0624 Ohm
Output Filter Capacitor	Cout, Co	39u, ESR = 35 mOhm	Aluminum Cap 50V Rated. 50SVF39M
Resonant Frequency		~5.04 kHz	

Circuit Representation of this table next page!



Circuit Representation

Majority of numerical-relevant results can be found in the upcoming “Performance Report” section. The subsequent sections might contain (only) formula / implementations (notably in the “calculation routine”). There are also two SPICE validation sections available later in the report

Appendix section contains all the MATLAB code.

%%%

Performance Report

Operation Condition: Vin = 32 V, PWR = 50W (Worst Efficiency Case)

Performance Evaluation

Condition:

Power [W]: 50

Input Voltage [V]: 32

Switching Frequency [Hz]: 265000

Input Filter and EMI Consideration

Maximum Allowed Filter Resonant Frequency (the design should be smaller) [rad/s]: 181160.8966

Filter Resonant Frequency [rad/s]: 62604.0189

Characteristics Filter Impedance [Ohm]: 1.5973

Equivalent Negative Converter Load [Ohm]: -20.48

Maximum Allowed FR Attenuation Factor [dB] @ switching frequency: -38.5345

See Attached Filter FR for attenuation evaluation @ switching frequency

Ripple Evaluations:

Output Voltage Ripple [Vpp], Limit 80mVpp: 0.014263

Input Current Ripple [A_pp], Limit 50mApp: 0.0029872

Inductor Specification:

Input Inductor Value [H]: 2.5515e-05

Input Inductor Magnetix Flux Density, limit 0.3T [T]: 0.018244

Input Inductor Current Density, limit 5000000 A/m² [A/m²]: 187093.7811

Output Inductor Value [H]: 2.5515e-05

Input Inductor Magnetix Flux Density, limit 0.3T [T]: 0.045718

Output Inductor Current Density, limit 5000000 A/m² [A/m²]: 468838.4838

Power Loss and Efficiency Evaluation:

FET Conduction Loss [W]: 0.18333

FET Switching Loss [W]: 0.49467

Diode Loss [W]: 0.88542

Input Filter Inductor Core Loss [W]: 3.1531e-06

Input Filter Inductor Conduction Loss DC [W]: 0.0056552

Input Filter Inductor Conduction Loss AC w/ Skin Depth [W]: 5.1899e-07

Output Filter Inductor Core Loss [W]: 1.3251

Output Filter Inductor Conduction Loss DC [W]: 0.025737

Output Filter Inductor Conduction Loss w/ Skin Depth [W]: 0.020219

Total Loss [W]: 2.9087

Given Loss Budget [W]: 5

Efficiency, Requirement > 0.9: 0.9412

Thermal Performance Evaluation:

Designed Effective FET Junction To Ambient Designed Thermal Impedance [C/W]: 27.254

FET Calculated Junction Temperature, Limit @ 150C [C]: 58.4782

Designed Effective Diode Junction To Ambient Designed Thermal Impedance [C/W]: 32.31

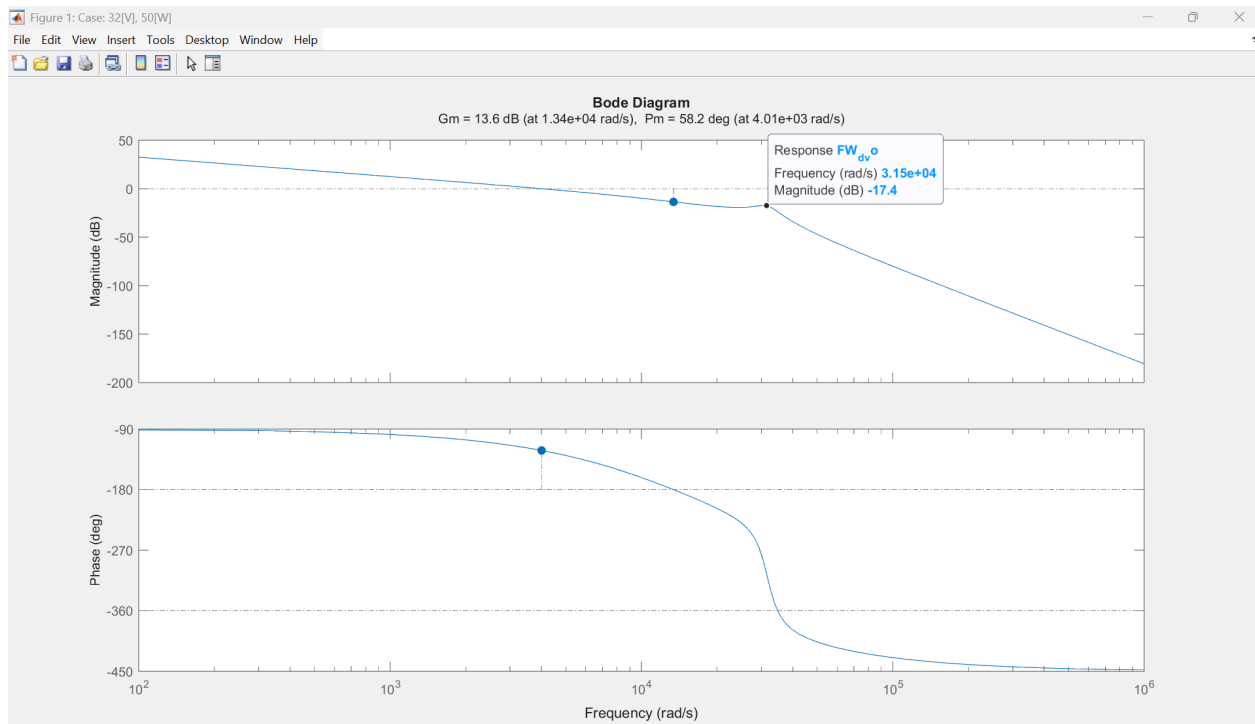
Diode Calculated Junction Temperature, Limit @ 150C [C]: 68.6078

Input Inductor Temperature (Assuming 50C Ambient) [C]: 50.1302

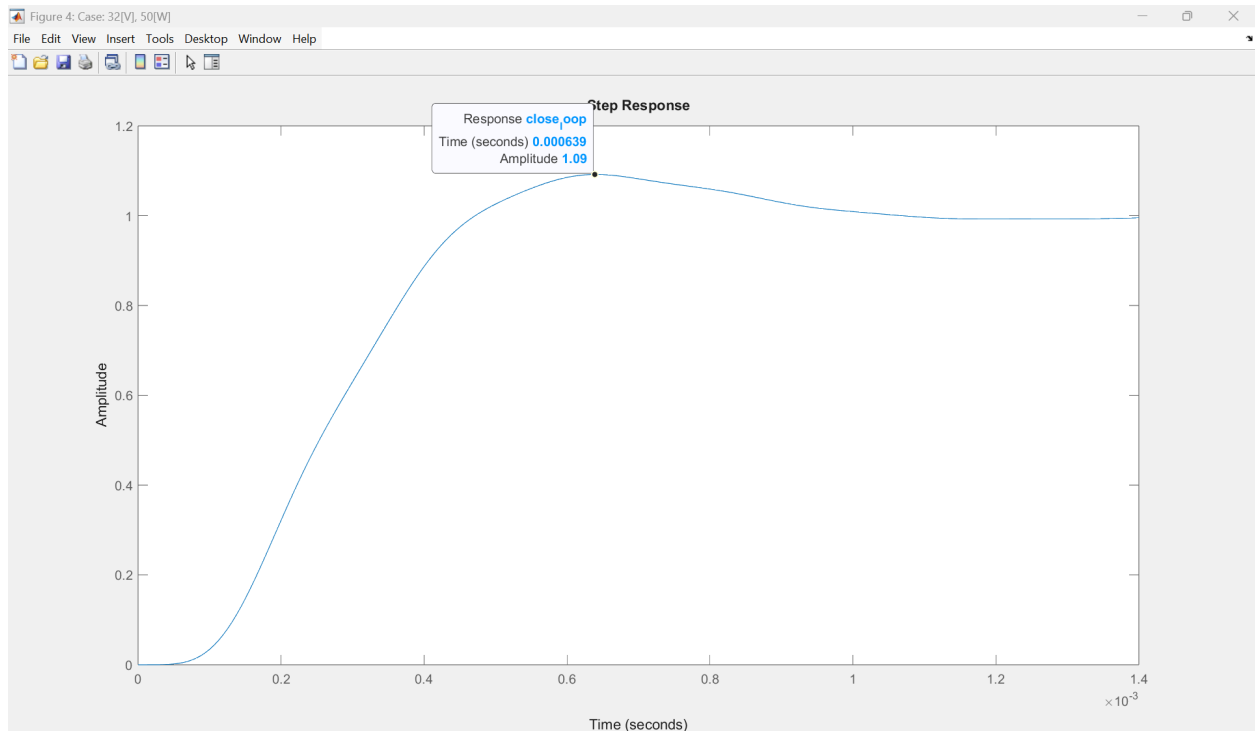
Output Inductor Temperature (Assuming 50C Ambient) [C]: 81.5334

Transient and Steady State Tolerance Performance:

See Controller Section of the Report

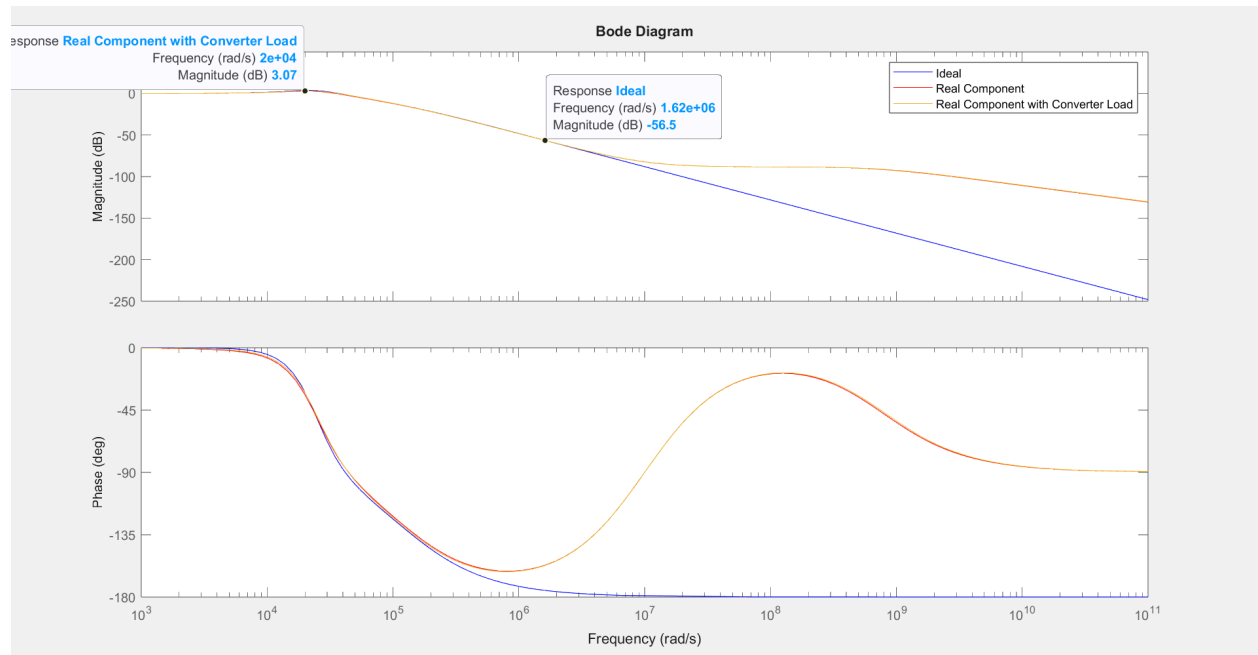


**Open Loop Transfer Function: Phase Margin: 58.2 > 45 degree, Gain Margin: 13.6 dB.
Resonant Gain Margin: 17.4dB. Theoretically infinite open loop DC gain.**



Close loop step response: Normalized Amplitude (need to be less than 120%)

See feedback analog simulation in a subsequent section!



EMI Input Filter Current Behavior

Peaking: 3.07dB

Attenuation @ 265 kHz (~1.67Mrad/s): -56.5 dB < -38.5dB

All 3 models behave the same @ switching frequencies

%%%

Performance Report

Operation Condition: $V_{in} = 32\text{ V}$, $PWR = 150\text{ W}$ (High FET and Diode Junction Temperature Case)

Performance Evaluation

Condition:

Power [W]: 150

Input Voltage [V]: 32

Switching Frequency [Hz]: 265000

Input Filter and EMI Consideration

Maximum Allowed Filter Resonant Frequency (the design should be smaller) [rad/s]: 104593.2924

Filter Resonant Frequency [rad/s]: 62604.0189

Characteristics Filter Impedance [Ohm]: 1.5973

Equivalent Negative Converter Load [Ohm]: -6.8267

Maximum Allowed FR Attenuation Factor [dB] @ switching frequency: -48.0769

See Attached Filter FR for attenuation evaluation @ switching frequency

Ripple Evaluations:

Output Voltage Ripple [Vpp], Limit 80mVpp: 0.014263

Input Current Ripple [A_pp], Limit 50mApp: 0.0089617

Inductor Specification:

Input Inductor Value [H]: 2.5515e-05

Input Inductor Magnetix Flux Density, limit 0.3T [T]: 0.054732

Input Inductor Current Density, limit 5000000 A/m² [A/m²]: 561281.3434

Output Inductor Value [H]: 2.5515e-05

Input Inductor Magnetix Flux Density, limit 0.3T [T]: 0.12341

Output Inductor Current Density, limit 5000000 A/m² [A/m²]: 1265582.0394

Power Loss and Efficiency Evaluation:

FET Conduction Loss [W]: 1.65

FET Switching Loss [W]: 1.484

Diode Loss [W]: 2.6562

Input Filter Inductor Core Loss [W]: 4.6525e-05

Input Filter Inductor Conduction Loss DC [W]: 0.050897

Input Filter Inductor Conduction Loss AC w/ Skin Depth [W]: 4.6709e-06

Output Filter Inductor Core Loss [W]: 1.3251

Output Filter Inductor Conduction Loss DC [W]: 0.23164

Output Filter Inductor Conduction Loss w/ Skin Depth [W]: 0.020219

Total Loss [W]: 7.1356

Given Loss Budget [W]: 15

Efficiency, Requirement > 0.9: 0.95055

Thermal Performance Evaluation:

Designed Effective FET Junction To Ambient Designed Thermal Impedance [C/W]: 27.254

FET Calculated Junction Temperature, Limit @ 150C [C]: 125.414

Designed Effective Diode Junction To Ambient Designed Thermal Impedance [C/W]: 32.31

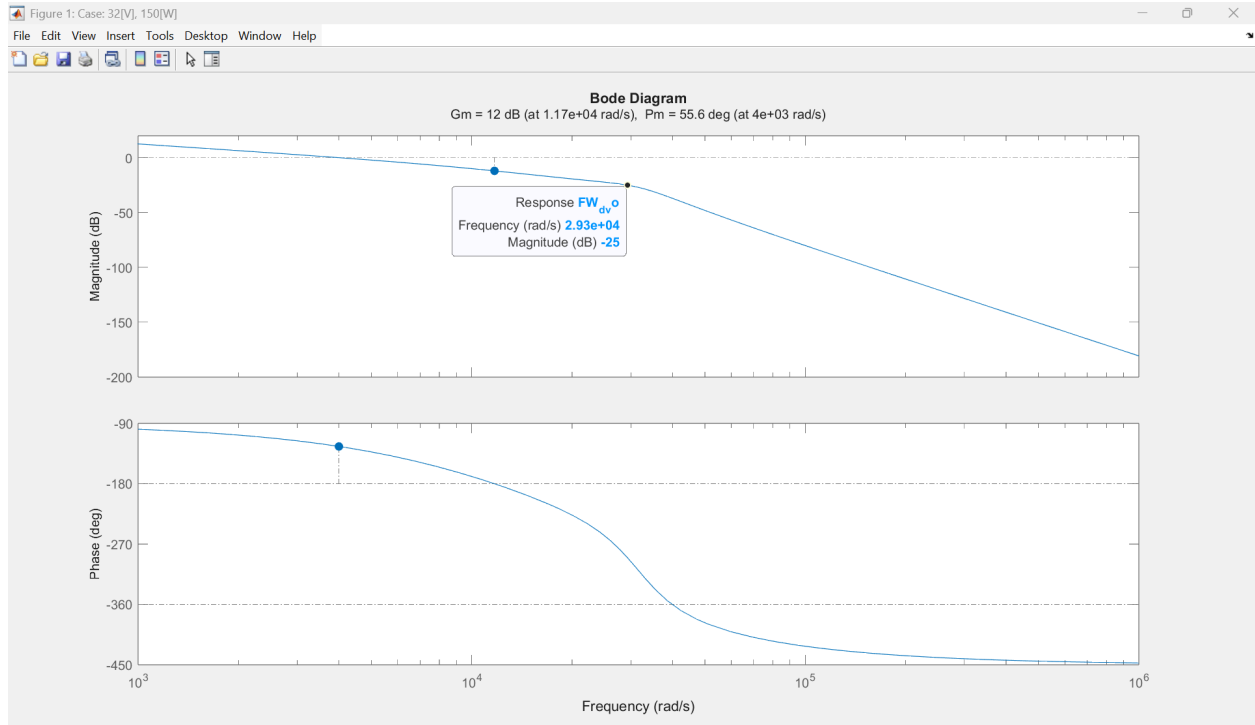
Diode Calculated Junction Temperature, Limit @ 150C [C]: 125.8234

Input Inductor Temperature (Assuming 50C Ambient) [C]: 51.1718

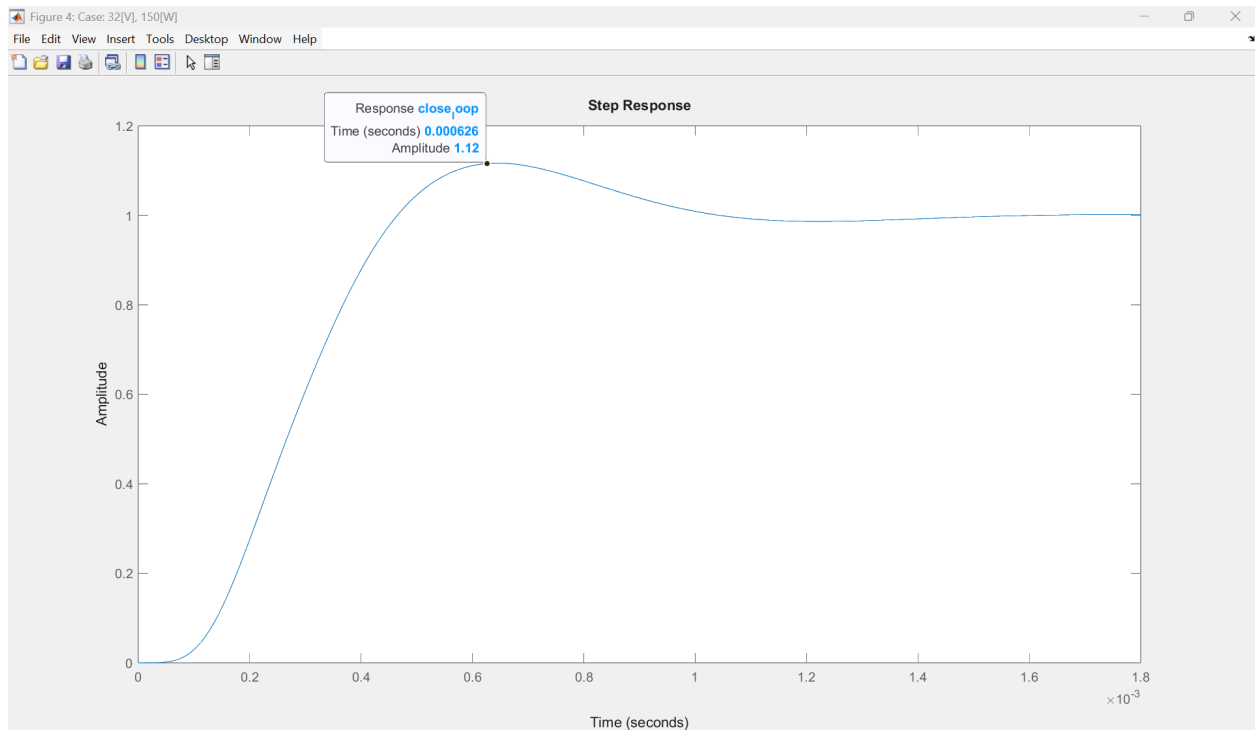
Output Inductor Temperature (Assuming 50C Ambient) [C]: 86.2691

Transient and Steady State Tolerance Performance:

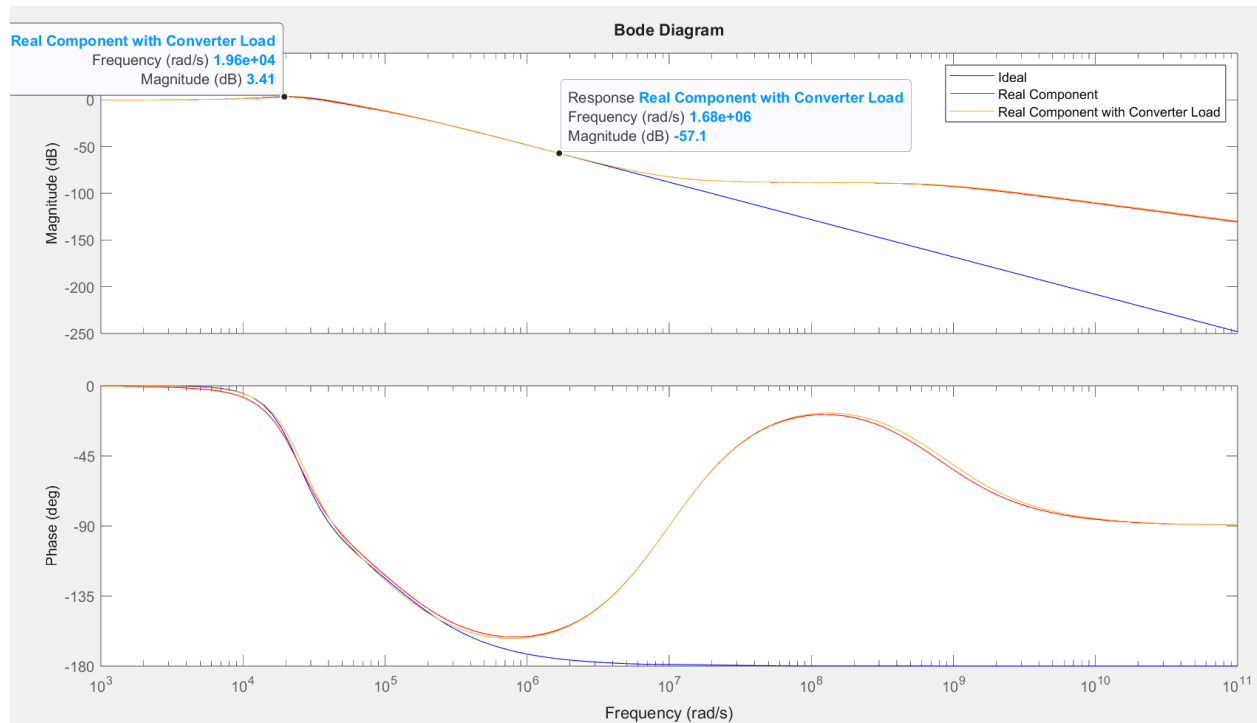
See Controller Section of the Report



**Open Loop Transfer Function: Phase Margin: 55.6 > 45 degree, Gain Margin: 12 dB.
Resonant Gain Margin: 25dB. Theoretically infinite open loop DC gain.**



Close loop step response: Normalized Amplitude



EMI Input Filter Current Behavior

Peaking: 3.41dB

Attenuation @ 265 kHz ($\sim 1.67 \text{ Mrad/s}$): -57.1 dB < -48.07dB

All 3 models behave the same @ switching frequencies

%%%

Performance Report

Operation Condition: $V_{in} = 22\text{ V}$, $PWR = 50\text{ W}$

Performance Evaluation

Condition:

Power [W]: 50

Input Voltage [V]: 22

Switching Frequency [Hz]: 265000

Input Filter and EMI Consideration

Maximum Allowed Filter Resonant Frequency (the design should be smaller) [rad/s]: 197039.3027

Filter Resonant Frequency [rad/s]: 62604.0189

Characteristics Filter Impedance [Ohm]: 1.5973

Equivalent Negative Converter Load [Ohm]: -9.68

Maximum Allowed FR Attenuation Factor [dB] @ switching frequency: -37.0749

See Attached Filter FR for attenuation evaluation @ switching frequency

Ripple Evaluations:

Output Voltage Ripple [Vpp], Limit 80mVpp: 0.0085424

Input Current Ripple [A_pp], Limit 50mApp: 0.0025252

Inductor Specification:

Input Inductor Value [H]: 2.5515e-05

Input Inductor Magnetix Flux Density, limit 0.3T [T]: 0.026516

Input Inductor Current Density, limit 5000000 A/m² [A/m²]: 271918.9106

Output Inductor Value [H]: 2.5515e-05

Input Inductor Magnetix Flux Density, limit 0.3T [T]: 0.042962

Output Inductor Current Density, limit 5000000 A/m² [A/m²]: 440576.4359

Power Loss and Efficiency Evaluation:

FET Conduction Loss [W]: 0.26667

FET Switching Loss [W]: 0.34008

Diode Loss [W]: 0.5303

Input Filter Inductor Core Loss [W]: 2.0891e-06

Input Filter Inductor Conduction Loss DC [W]: 0.011965

Input Filter Inductor Conduction Loss AC w/ Skin Depth [W]: 3.7085e-07

Output Filter Inductor Core Loss [W]: 0.37741

Output Filter Inductor Conduction Loss DC [W]: 0.025737

Output Filter Inductor Conduction Loss w/ Skin Depth [W]: 0.007253

Total Loss [W]: 1.5217

Given Loss Budget [W]: 5

Efficiency, Requirement > 0.9: 0.96881

Thermal Performance Evaluation:

Designed Effective FET Junction To Ambient Designed Thermal Impedance [C/W]: 27.254

FET Calculated Junction Temperature, Limit @ 150C [C]: 56.5364

Designed Effective Diode Junction To Ambient Designed Thermal Impedance [C/W]: 32.31

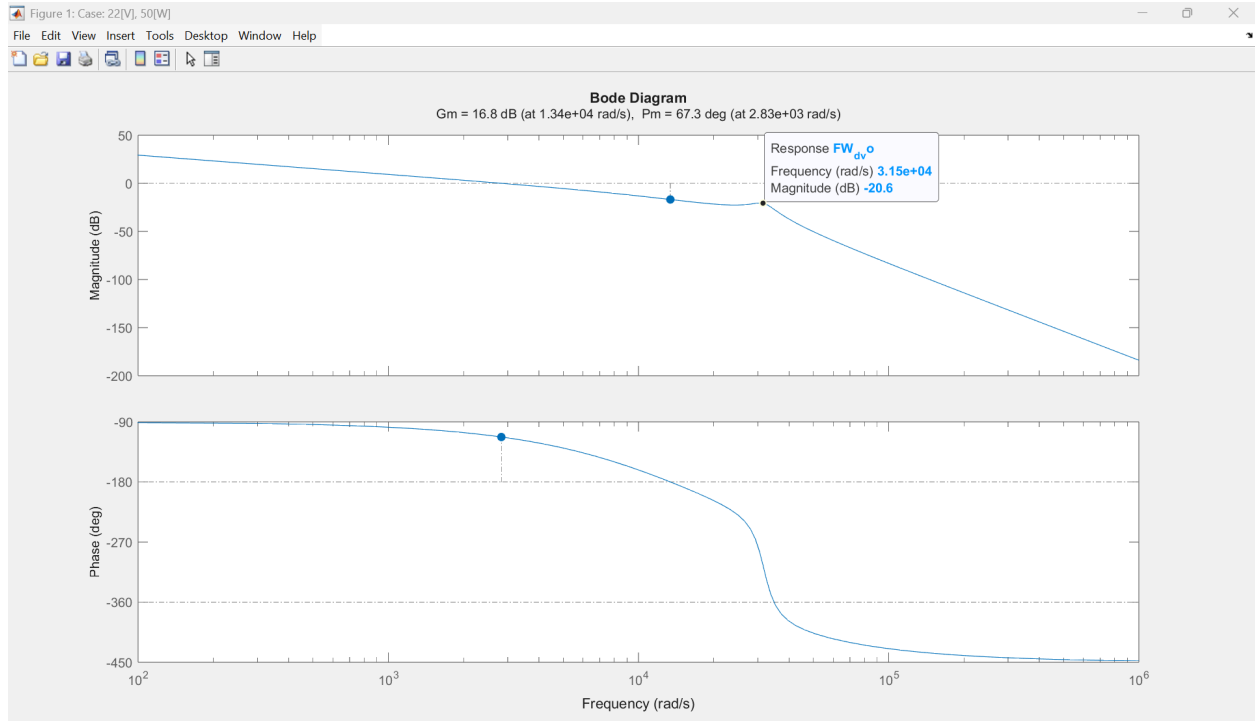
Diode Calculated Junction Temperature, Limit @ 150C [C]: 57.1341

Input Inductor Temperature (Assuming 50C Ambient) [C]: 50.2752

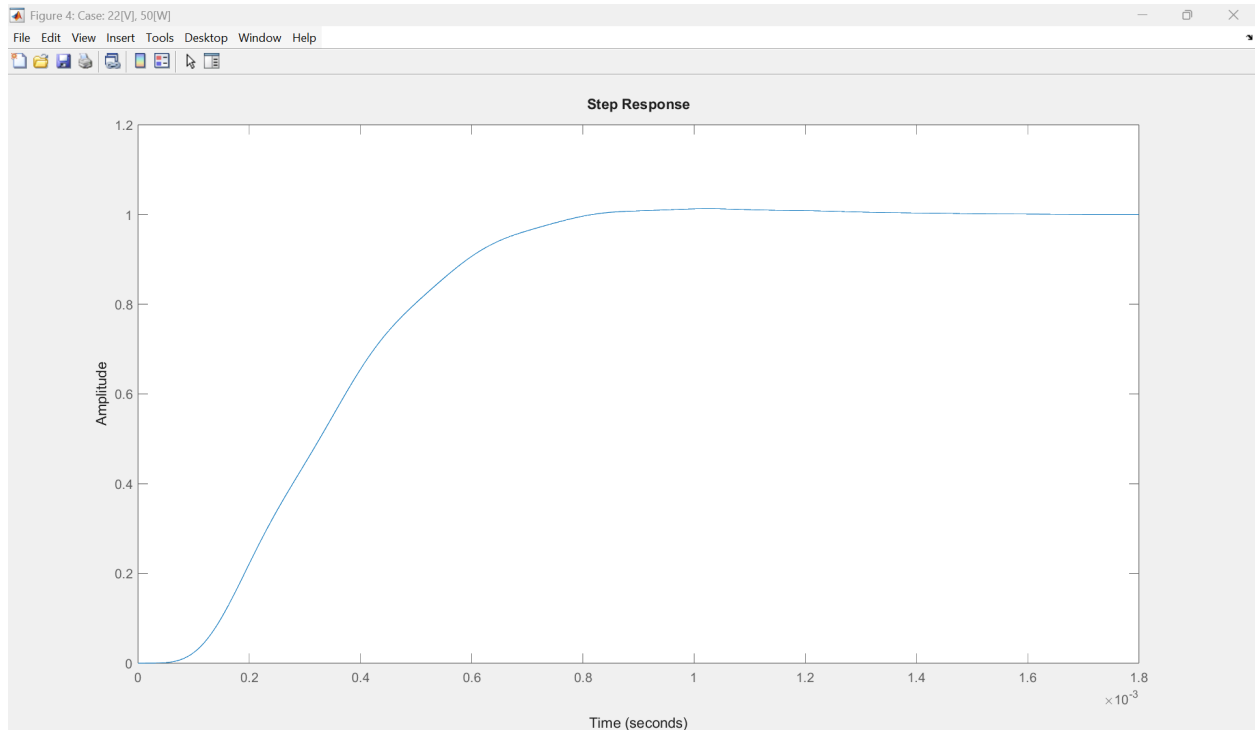
Output Inductor Temperature (Assuming 50C Ambient) [C]: 59.4391

Transient and Steady State Tolerance Performance:

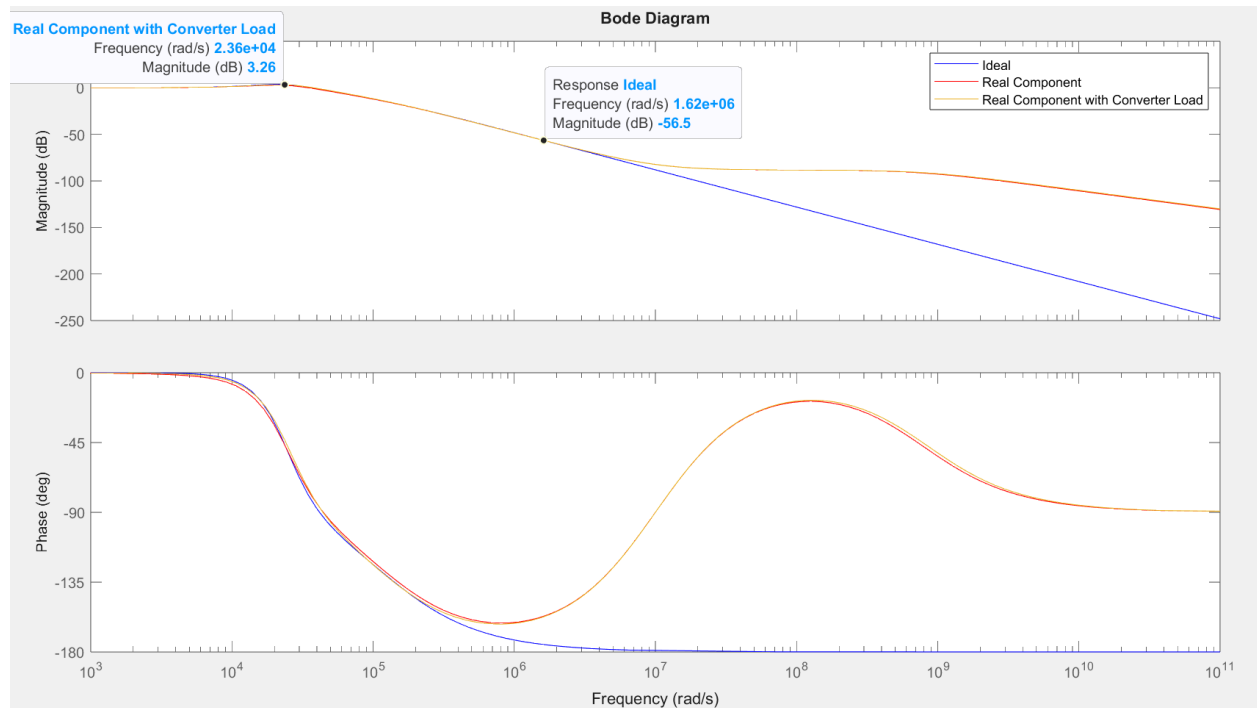
See Controller Section of the Report



**Open Loop Transfer Function: Phase Margin: 67.3 > 45 degree, Gain Margin: 16.8 dB.
Resonant Gain Margin: 20.6dB**



Close loop step response: Normalized Amplitude



EMI Input Filter Current Behavior

Peaking: 3.26dB

Attenuation @ 265 kHz (~ 1.67 Mrad/s): -56.5 dB < -37.07dB

All 3 models behave the same @ switching frequencies

%%%

Performance Report

Operation Condition: Vin = 22 V, PWR = 150W (Heavy Negative Load)

Performance Evaluation

Condition:

Power [W]: 150

Input Voltage [V]: 22

Switching Frequency [Hz]: 265000

Input Filter and EMI Consideration

Maximum Allowed Filter Resonant Frequency (the design should be smaller) [rad/s]: 113760.6945

Filter Resonant Frequency [rad/s]: 62604.0189

Characteristics Filter Impedance [Ohm]: 1.5973

Equivalent Negative Converter Load [Ohm]: -3.2267

Maximum Allowed FR Attenuation Factor [dB] @ switching frequency: -46.6173

See Attached Filter FR for attenuation evaluation @ switching frequency

Ripple Evaluations:

Output Voltage Ripple [Vpp], Limit 80mVpp: 0.0085424

Input Current Ripple [A_pp], Limit 50mApp: 0.0075756

Inductor Specification:

Input Inductor Value [H]: 2.5515e-05

Input Inductor Magnetix Flux Density, limit 0.3T [T]: 0.079547

Input Inductor Current Density, limit 5000000 A/m^2 [A/m^2]: 815756.7318

Output Inductor Value [H]: 2.5515e-05

Input Inductor Magnetix Flux Density, limit 0.3T [T]: 0.12065

Output Inductor Current Density, limit 5000000 A/m^2 [A/m^2]: 1237319.9916

Power Loss and Efficiency Evaluation:

FET Conduction Loss [W]: 2.4

FET Switching Loss [W]: 1.0203

Diode Loss [W]: 1.5909

Input Filter Inductor Core Loss [W]: 3.0825e-05

Input Filter Inductor Conduction Loss DC [W]: 0.10768

Input Filter Inductor Conduction Loss AC w/ Skin Depth [W]: 3.3377e-06

Output Filter Inductor Core Loss [W]: 0.37741

Output Filter Inductor Conduction Loss DC [W]: 0.23164

Output Filter Inductor Conduction Loss w/ Skin Depth [W]: 0.007253

Total Loss [W]: 5.3959

Given Loss Budget [W]: 15

Efficiency, Requirement > 0.9: 0.96177

Thermal Performance Evaluation:

Designed Effective FET Junction To Ambient Designed Thermal Impedance [C/W]: 27.254

FET Calculated Junction Temperature, Limit @ 150C [C]: 133.2155

Designed Effective Diode Junction To Ambient Designed Thermal Impedance [C/W]: 32.31

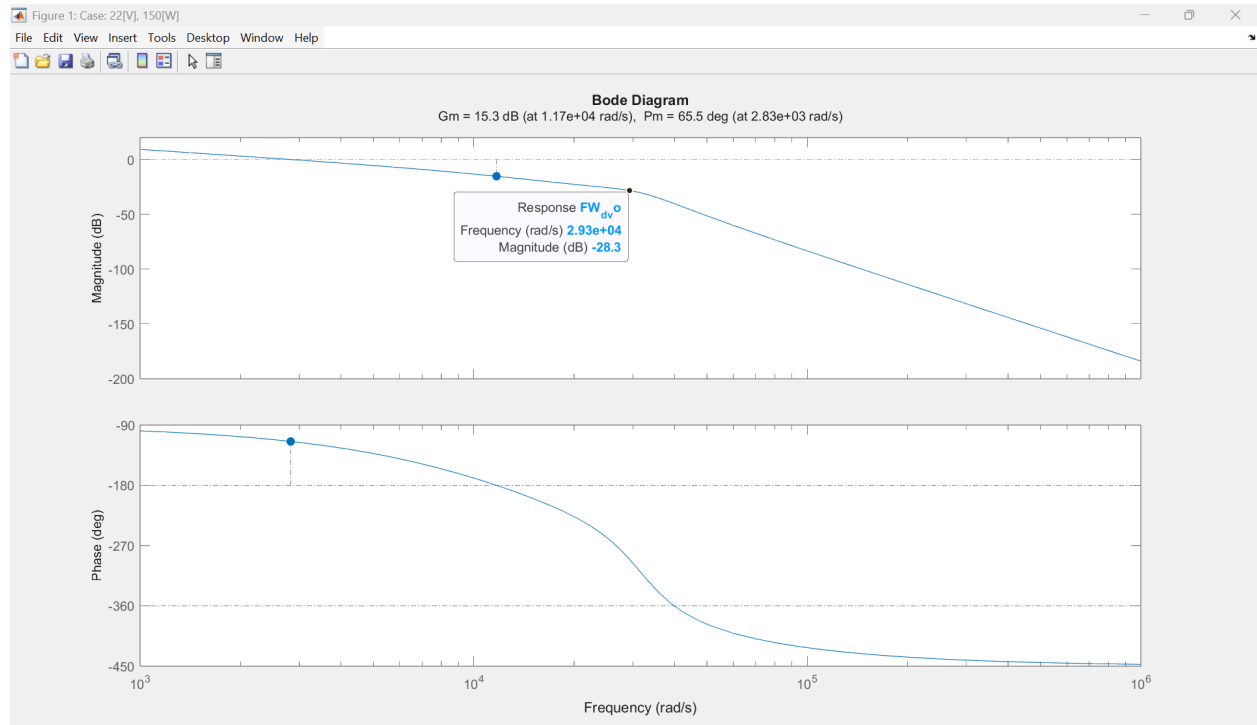
Diode Calculated Junction Temperature, Limit @ 150C [C]: 91.4023

Input Inductor Temperature (Assuming 50C Ambient) [C]: 52.4775

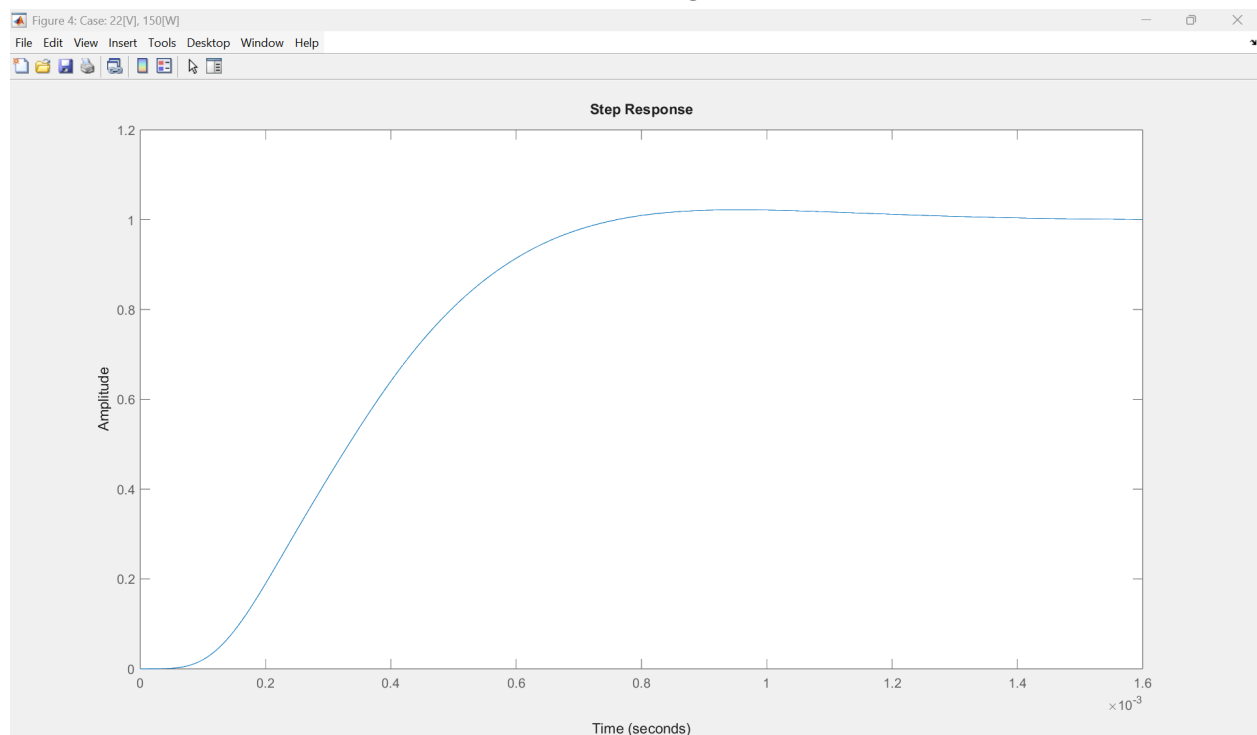
Output Inductor Temperature (Assuming 50C Ambient) [C]: 64.1748

Transient and Steady State Tolerance Performance:

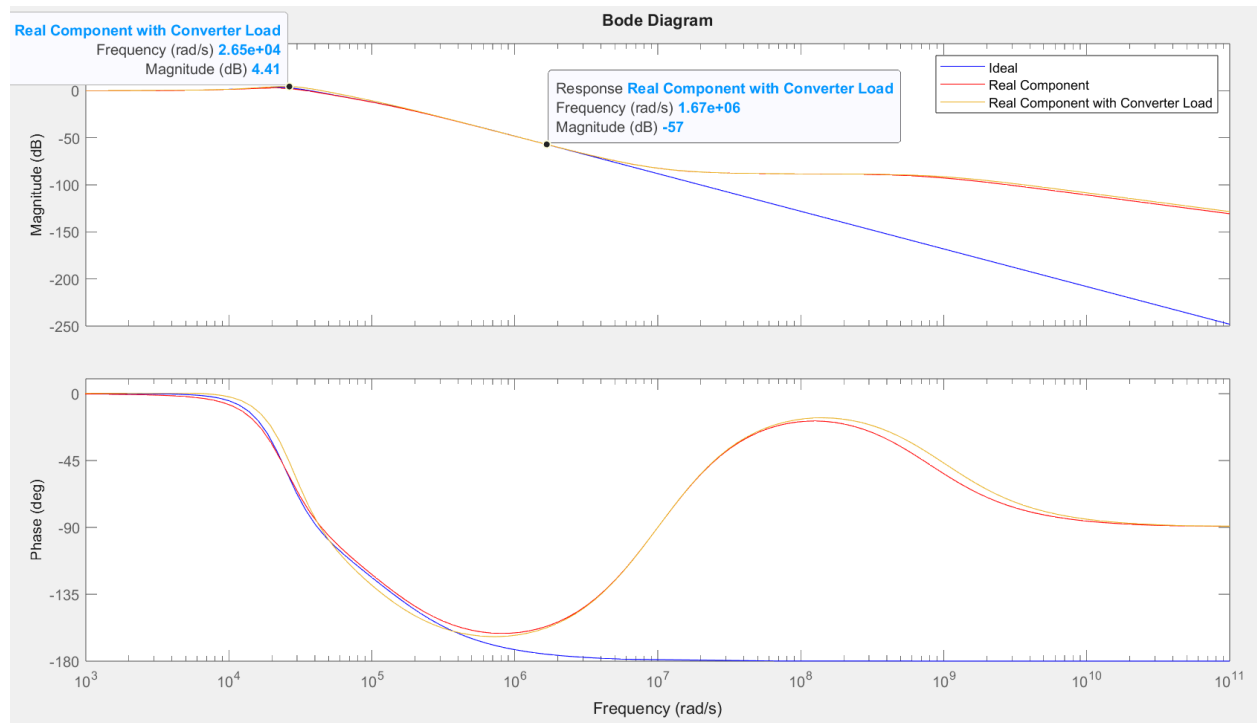
See Controller Section of the Report



**Open Loop Transfer Function: Phase Margin: 65.5 > 45 degree, Gain Margin: 15.3 dB.
Resonant Gain Margin: 28.3dB**



Close loop step response: Normalized Amplitude.



EMI Input Filter Current Behavior

Peaking: 4.41 dB (Highest of all cases)

Attenuation @ 265 kHz ($\sim 1.67 \text{Mrad/s}$): -57 dB < -46.06 dB

All 3 models behave the same @ switching frequencies

%%%

Essential Derived Starter Parameters

These are some of the parameters that can be derived given the requirement, which will be used at various design stages (I_{in} , Duty Cycle, etc.)

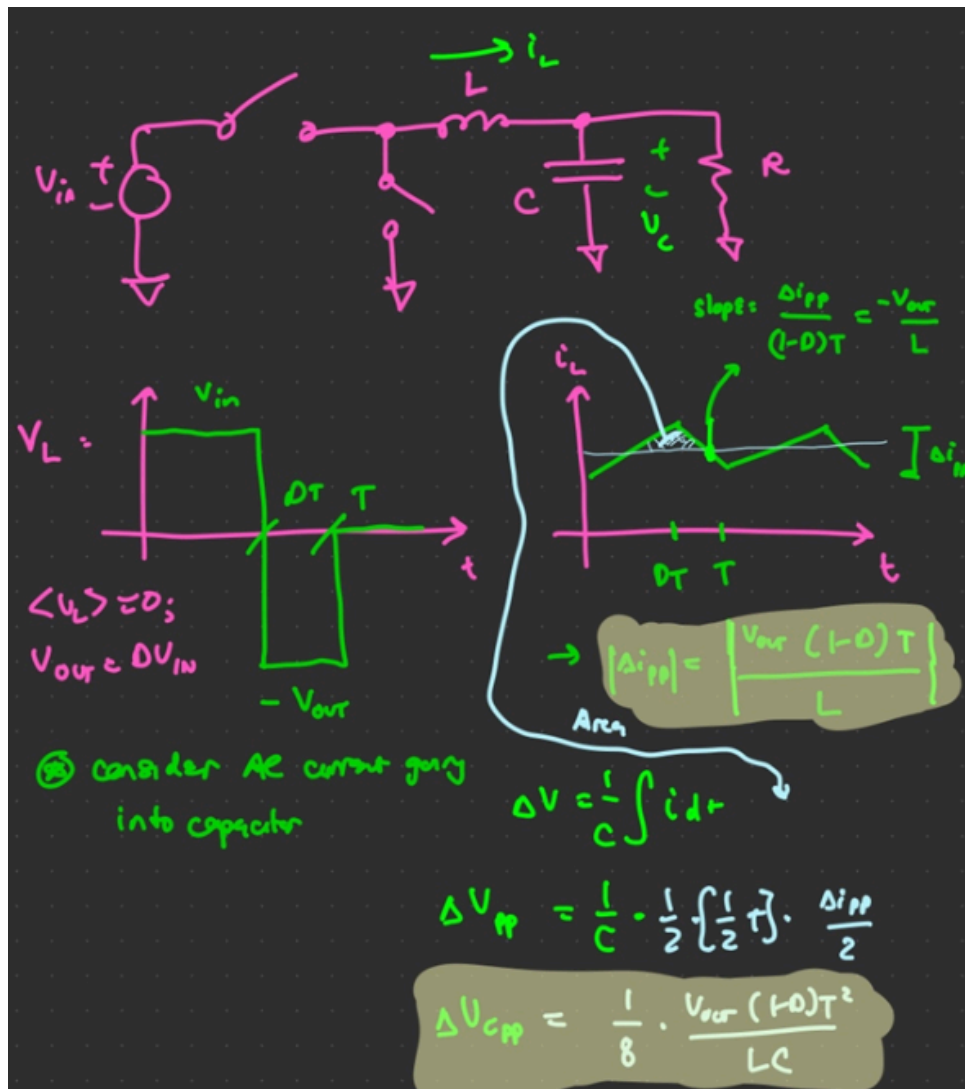
85	%% Derived Parameters	
86	$D_{ccm} = V_{out} ./ V_{in};$	
87	$I_{out} = PWR ./ V_{out};$	
88	$R_{conv_neg} = -(V_{in})^2 / (PWR)$ % $r = P / (i_{in}^2) = P / (P/v_{in})^2 = v_{in}^2 / P$.	Lowest power highest Vin is worst case
89	$I_{in} = PWR ./ V_{in};$	
90	%% Derived Parameters - EMI Target	
91	$isource_fundamental = I_{out} ./ \pi * \sqrt{2 * 2 * \cos(2 * \pi * D_{ccm})};$	
92	$win_target = \sqrt{lim_ippin_rip * 1/2 .* 1 ./ (isource_fundamental) * (2 * \pi * f_{sw})^2};$	
93	$win_target_conv_factor = \sqrt{lim_ippin_rip * 1/2 .* 1 ./ (isource_fundamental)};$	%smaller is harder (require bigger component)
94		

- Duty Cycle: $V_{out} = D V_{in}$
 - Assume CCM Condition
- I_{in} : Use Power Balance ($V_{in} * I_{in} = V_{out} * I_{out}$) to solve for I_{in}
- I_{out} : Power / V_{out}
- EMI Related Metrics
 - Negative Equivalent Resistance
 - Fundamental Harmonic of the switching current seen by FET
 - “High Level” of the associated pulse waveform is I_{out}
 - Target frequency “win_target” derive the input filter resonant frequency value that meets the input ripple current target
 - This helps guide the input filter sizing. See EMI section for more detailed design

%%%

Output Voltage Ripple and Output Inductor & Capacitor Sizing

Output Inductor Current Ripple and Output Voltage Ripple



```

119 %% Output Ripple Derivation
120 % OUTPUT
121 i_Lo_rip_pp = Vout.*(1-Dccm).*Tsw./Lo;
122 % Reminder: for ripple model: Constant Current load is open! -> i ripple
123 % goes into C rippl = i = C dv dt -> integral (i) / C = dv. Inductor Current ripple
124 % has triangle shape.
125 v_Co_rip_pp = Vout.*(1-Dccm).*Tsw^2/(8*Lo*Co);
126

```

$$i_{oripp} = \frac{V_{out}(1-D) \cdot T}{L_o}, \quad v_{oripp} = \frac{V_{out} \cdot (1-D) \cdot T^2}{8 \cdot L_o \cdot C_o}$$

“v_{Co_rip_pp}” is the output voltage ripple term of interest. L and C need to be sufficiently “large” such that the ripple voltage is less than 80mVpp. Converter is assumed to be working in CCM, which is true for all 4 cases

For my design, the following parameters are chosen:

- T_{sw} = 1/265000 s
- L_o = 25.5uH
 - See “Inductor Design” Section for Specifications
- C_o = 39uF - **50SVF39M**
 - Aluminum Capacitor is used for its storage. Rated @ 50V.
 - ESR = 0.035 Ohm

%%%

- **EMI Consideration - Input Filter Design**

- To start with designing the EMI filter, the fundamental harmonic of current can be calculated. To obtain a desired input filter inductor (L_f) and capacitor (C_f) should be sized such that its resonant frequency is lower than the targeted filter frequency, which is calculated by considering the fundamental harmonics of FET switching current, input current ripple requirement, as shown in the calculation routine below. EMI_TF_dB_attenuation_factor denotes current ratio required to meet specification in dB, and will be used for FR validation later:

```
% Derived Parameters - EMI Target
isource_fundamental = I_out./pi.*sqrt(2-2*cos(2*pi.*Dccm));
% Maximum Filter Characteristic Frequency. Smaller Value Require Larger L/C
EMI_TF_dB_attenuation_factor = mag2db(lim_ippin_rip*(1/2) / isource_fundamental);
win_target = sqrt(lim_ippin_rip*1/2 .* 1./(isource_fundamental) * (2*pi*fsw)^2);
win_target_conv_factor = sqrt(lim_ippin_rip*1/2 .* 1./(isource_fundamental));
```

- Characteristics Impedance of the filter network is also checked against the equivalent negative impedance presented by the converter ($Z_{char} < R_{neg_conv}$). (See performance report for numerical values). Degradation in damping around resonant frequency will be observed if this condition is not satisfied.
- After the components are selected for the first pass, a more involved calculation routine involves plotting FR - including all parasitics & Rd-Cd network on the filter performance. As an estimation, ESL is set to be at 1nH. The routine is defined as follow:

```
177 % Input Filter FR
178 s=tf('s');
179 Z_Cin_real = 1/(Cin*s) + Resr_Cin + Lesr_Cin*s;
180 Z_Lin_real = Lin*s + Resr_Lin;
181 Zf_char = sqrt(Lin/Cin);
182 zeta_in = Cd / Cin;
183 f_f_input = 1/(2*pi*sqrt(Lin*Cin));
184 % formula to find an optimal Rd resistor value
185 Rd = Zf_char * sqrt((2+zeta_in)*(4+3*zeta_in)/((2*zeta_in^2)*(4+zeta_in)));
186
187 Z_d = 1/(Cd*s) + Rd;
188
189 H_s_ideal = (1 + Rd*Cd*s)/(1 + Rd*Cd*s + Lin*(Cd+Cin)*s^2 + Lin*Rd*Cd*Cin*s^3);
190 H_s_real = ((1/Z_d + 1/Z_Cin_real + 1/(Z_Lin_real))^(-1)) / (Z_Lin_real);
191
192 H_s_with_converter_load = ( (1/Z_d + 1/Z_Cin_real + 1/(Z_Lin_real) + 1/(R_conv_neg))^(-1) ) / (Z_Lin_real);
193
194 figure();
195 bode(H_s_ideal, "b");
196 hold on
197 bode(H_s_real, "r");
198 bode(H_s_with_converter_load)
199 legend("Ideal", "Real Component", "Real Component with Converter Load")
200
```

- The “H_s_ideal” transfer function depicts current ratio for ideal component, while “H_s_real” transfer function depicts the current ratio as real component impedance is taken into account. There’s also “H_s_with_converter_load” that takes into account the effect of the converter. These transfer functions are plotted and shown in the performance report section - the magnitude at this transfer function at switching frequency is of interest.
 - It turns out that at switching frequency, all 3 transfer functions behave pretty much identical for all 4 boundary cases (refer to performance report)
 - If I operated my design @ higher frequency, this would not be true
- Optimal R_d is calculated from a chosen L_f , C_f and C_d value.
- The transfer function is then evaluated @ switching frequency:
 - Notably, it’s checked against the “EMI_TF_dB_attenuation_factor”, which is the maximum attenuation factor allowed. The transfer function should offer a more negative value. (If not, the design won’t meet input current ripple requirements. This is important as the transfer function incorporates parasitics dynamics.)
 - The transfer function is plotted in the performance report section.
- Chosen Components:
 - Inductor (L_f): 25.5 μH
 - See Inductor Design Section for Details
 - Filter Capacitor C_f : 10 μF (KTS500B106M55N0T00)
 - Multilayer Ceramics, Good for high Frequency Performance. 50V rated.
 - 2A rms ripple current rating
 - C_d : 68 μF
 - Aluminum Capacitor - 50SVF68M, 50V rated
 - ESR = 0.025 Ω
 - R_d optimal value is then calculated: 0.74 Ω
 - Thus, I need to add an extra ~0.715 Ω resistor to reach the optimal resistance value.

%%%

Semiconductor Selection (FET and Diode)

- **FET Selection: NTMFS5844NL**
 - **$V_{DS_breakdown} = 60V$, $ID_{max} = 60A$**
 - @ 15V / 150W condition, this diode should be within voltage and current operating range with sufficient margin
 - **R_{dson} : MAX 16mOhm**
 - Use 16mOhm x 2.2 to take into account increase in resistance when device heating up

SWITCHING CHARACTERISTICS (Note 6)

Turn-On Delay Time	$t_{d(ON)}$	$V_{GS} = 4.5\text{ V}, V_{DS} = 48\text{ V},$ $I_D = 10\text{ A}, R_G = 2.5\ \Omega$	12		ns
Rise Time	t_r		25		
Turn-Off Delay Time	$t_{d(OFF)}$		20		
Fall Time	t_f		10		

- This device has a relatively lower rise and fall time comparatively to other provided devices (which has around ~50-60ns in total rise and fall time)
 - Useful if I want for higher switching time

CHARGES, CAPACITANCES & GATE RESISTANCE

Input Capacitance	C_{ISS}	$V_{GS} = 0\text{ V}, f = 1\text{ MHz}, V_{DS} = 25\text{ V}$	1460		pF
Output Capacitance	C_{OSS}		150		
Reverse Transfer Capacitance	C_{RSS}		96		

- The gate driver should be able to sink and source enough current to drive a 1460pF load (~1.5nF) load @ 250 kHz

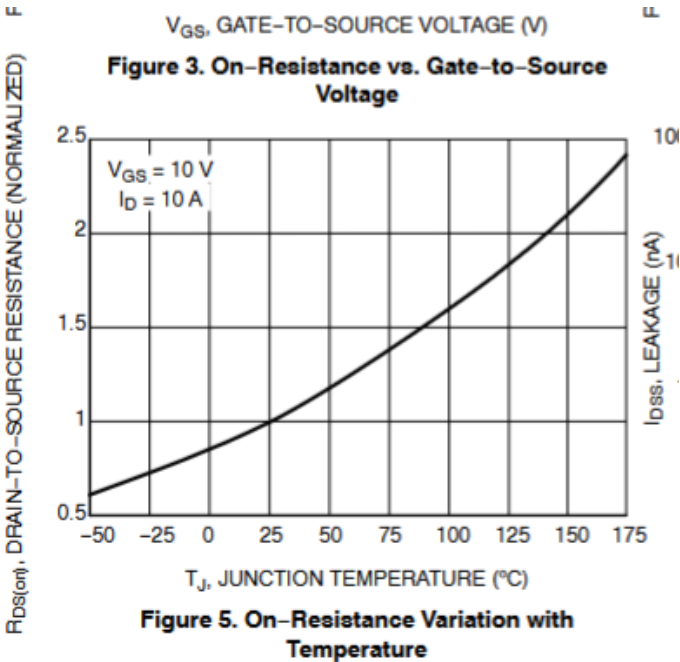
- **Diode Selection: PMEG045V150EPD**

- 45V and 15A Limit
 - @ 15V / 150W condition, this diode should be within voltage and current operating range with ample margin (~10A)
- Maximum Forward Voltage Drop: 490mV
 - Round up to 500 mV for thermal dissipation calculation
 - Thermal Impedance - Junction to Solder point: 3K/W

%%%

Semiconductor Losses (FET and Diode)

- **FET:**
 - Conduction Losses
 - $R_{ds(on)} \cdot I_{rms}^2 \cdot D$ (FET acts like a switch in triode region with a resistance $R_{ds(on)}$)
 - Note that $R_{ds(on)}$ has a factor of x2.2 to take into account temperature effect



- Switching Losses:
 - Dependent on rise and fall, and other operating variables (see implementation below). Both rise and fall duration contribute to loss.
 - For a given Voltage, Current, and switch characteristics, every switching cycle includes a “fixed” amount of loss. Faster Switching Frequency means switching losses occur more often, leading to higher switching losses. See Calculation Routine below for formula.

SWITCHING CHARACTERISTICS (Note 6)

Turn-On Delay Time	$t_{d(ON)}$	$V_{GS} = 4.5\text{ V}, V_{DS} = 48\text{ V},$ $I_D = 10\text{ A}, R_G = 2.5\ \Omega$		12		ns
Rise Time	t_r			25		
Turn-Off Delay Time	$t_{d(OFF)}$			20		
Fall Time	t_f			10		

- **Diode Losses:**
 - I_{rms} * Forward Voltage of Diode
 - Set at 0.5V
- **Calculation Routine for Semiconductor Losses:**
 - For this FET: $t_r = 25ns$, $t_f = 10ns$.
 - This option seems to have the lowest combined t_f+t_r compared to other options, which was why I picked this to allow me to have a higher switching frequency.

```

207 %% Semiconductor Losses
208 %FET & Diode Characteristics
209 t_r = 25*10^(-09);
210 t_f = 10*10^(-09);
211 Rdson = 0.016 * 2.2; %take into account x2.2 power dissipation
212 V_diode = 0.5;
213
214 % FET and Diode Losses
215 P_cond_FET = max(I_out)^2 * max(Dccm) * Rdson; % Conduction Loss:
216 P_sw_FET = 1/2*(t_f+t_r) * max(I_out) * max(Vin)*fsw; % switching loss
217 P_sw_total = P_cond_FET + P_sw_FET;
218
219 P_diode = max(V_diode*I_out.*(1-Dccm));
220

```

See the “Thermal Design for Semiconductor Components” for Thermal Design

%%%

Input and Output Inductors Design

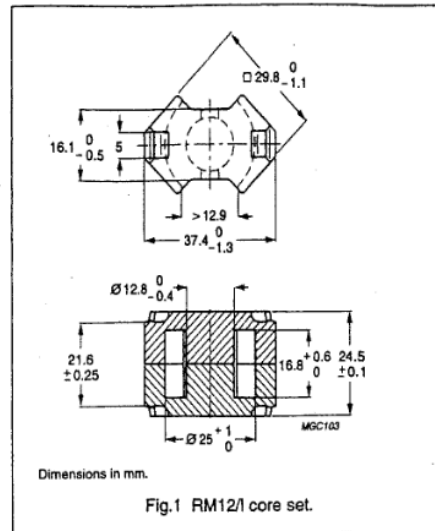
RM cores and accessories

RM12/I

CORE SETS

Effective core parameters

SYMBOL	PARAMETER	VALUE	UNIT
$\Sigma(l/A)$	core factor (C1)	0.388	mm ⁻¹
V_e	effective volume	8340	mm ³
l_e	effective length	56.6	mm
A_e	effective area	146	mm ²
A_{min}	minimum area	125	mm ²
m	mass of set	≈45	g



GRADE	A_L (nH)	μ_e	AIR GAP (μ m)	TYPE NUMBER
3F3	160 ±3%	≈49	≈1400	RM12/I-3F3-A160
	250 ±3%	≈77	≈800	RM12/I-3F3-A250
	315 ±5%	≈97	≈550	RM12/I-3F3-A315

Winding data for RM12/I coil former (DIL)

NUMBER OF SECTIONS	AVERAGE LENGTH OF TURN (mm)	WINDING AREA (mm ²)	WINDING WIDTH (mm)	TYPE NUMBER
1	61	75.0	14.3	CPV-RM12/I-1S-12PD

- Both input and output inductors are made the same way!
 - 9 Turns, 10 AWG wire ($D=2.88$ mm, Area ~ 5.3 mm²)
 - $75\text{mm}^2 / 5.3\text{mm}^2 \sim 15 > 9$
 - $\rightarrow$ Should fit and have enough margin even taking into account packing density
 - $A_L = 315$ nH, this leads to $L_{in} = L_{out} = 25.5$ uH
 - Gapped core for better energy storage and saturation behavior.

Calculation Routine:

```
28 i_L_max = i_avg + i_Lo_rip_pp/2;
29 MMF = i_L_max * N_turns;
30 MMF_rippl = max(i_Lo_rip_pp/2) * N_turns;
31
32 Reluc_core = L_core_eff/(mu_o*mu_rel*A_core_eff);
33 Reluc_gap = L_Airgap/(mu_o*A_core_eff);
34 Reluc_eff = Reluc_core + Reluc_gap;
35
36 Flux = MMF/Reluc_eff;
37 Flux_AC = MMF_rippl / Reluc_eff;
38 Flux_Linkage = N_turns*Flux;
39 Inductance = N_turns^2/Reluc_eff;
40 Inductance_method2 = A_L*N_turns^2;
41 B_core = Flux/A_core_min;
42 B_AC_rippl = Flux_AC / A_core_min;
43
44 I_density_max = (i_avg+i_Lo_rip_pp/2)/(pi * D_wire^2 / 4);
45
```

Explanation:

- Maximum current (and subsequently is used for calculating magnetic flux density)
 - For input stage, this is input DC current + ½ peak to peak ripple current
 - Input DC Current: PWR / V_{in}
 - For output stage, this is output DC current + ½ peak to peak ripple current
 - Output DC Current: PWR / V_{out}
 - Given MMF is then calculated
 - There's also an MMF_ripple term that's used for Core loss calculation later.
- Inductance can be computed: $L = A_L \times N^2$ (labelled “inductance_method2 and this is what I used for calculation”)
- Effective Reluctance is calculated by summing core reluctance and air gap reluctance
- Flux, Flux Linkage, Inductance and Magnetic Flux Density (B) is calculated
 - B needs to be less than 0.3T for saturation
 - I decide to size magnetic component for extra margin such that magnetic flux density $B < 0.2T$
 - Minimum Area is used for this calculation.
- Conductor current density is also checked to make sure the conductor has sufficient diameter and can carry the current (need to be less than $500A / cm^2$ or $5000000 A / m^2$)
- AC Flux and Flux density are used for core loss calculation. See the “Inductor Related Losses” section.

%%%

Inductor Related Losses Calculation. Temperature Rise Calculation for the inductor

Calculation Routine: “P_core_out”, “P_AC” and “P_DC” are among variables of interest

```
47 C_mu = 2.5*10^(-4);
48 alpha_core = 1.63;
49 beta_core = 2.45;
50 P_core_out = C_mu * (fsw ^ alpha_core) * ((B_AC_ripple*1/2)^beta_core) * V_core ;
51
52 res_per_length = 3.277*10^(-3); % 10 AWG
53 R_DC = D_core*pi*res_per_length*N_turns;
54 delta_skin = 0.00013;
55 W_delta_ratio = D_wire/delta_skin;
56
57 R_AC = R_DC * W_delta_ratio * (sinh(2*W_delta_ratio) + sin(2*W_delta_ratio))/(cosh(2*W_delta_ratio) - cos(2*W_delta_ratio))
58 fprintf("R_AC is" + R_AC + "\n")
59 P_DC = i_avg^2 * R_DC;
60 P_AC = (i_Lo_rip_pp/2)^2 * R_AC;
61
62 end
```

- DC Winding Loss:
 - Estimated: Winding Resistance x I_{DC}^2
- Skin Effect - AC Loss
 - @ 250 kHz, the skin depth in copper is less than the diameter of 10 AWG wire, and thus skin depth effect should be taken into account
 - Loss is calculated by finding equivalent “R_AC” value resistance, and combining it with AC current in the inductor
 - Usage the amplitude value ($i_{Lo_rip_pp}/2$) is conservative. RMS would be a more accurate estimation.
- Core Loss:
 - Using the provided empirical formula, and B_AC value mentioned in the previous section
- Total Inductor Loss is the sum of the aforementioned terms, is then used to calculate the temperature rise in the inductor (and making sure it does not exceed 100 C (assuming 50C starting point)
 - The chosen RM12 core has Thermal Impedance (junction to ambient) of 23 C/W
 - Temperature of inductors are included in the performance report

% Input Inductor

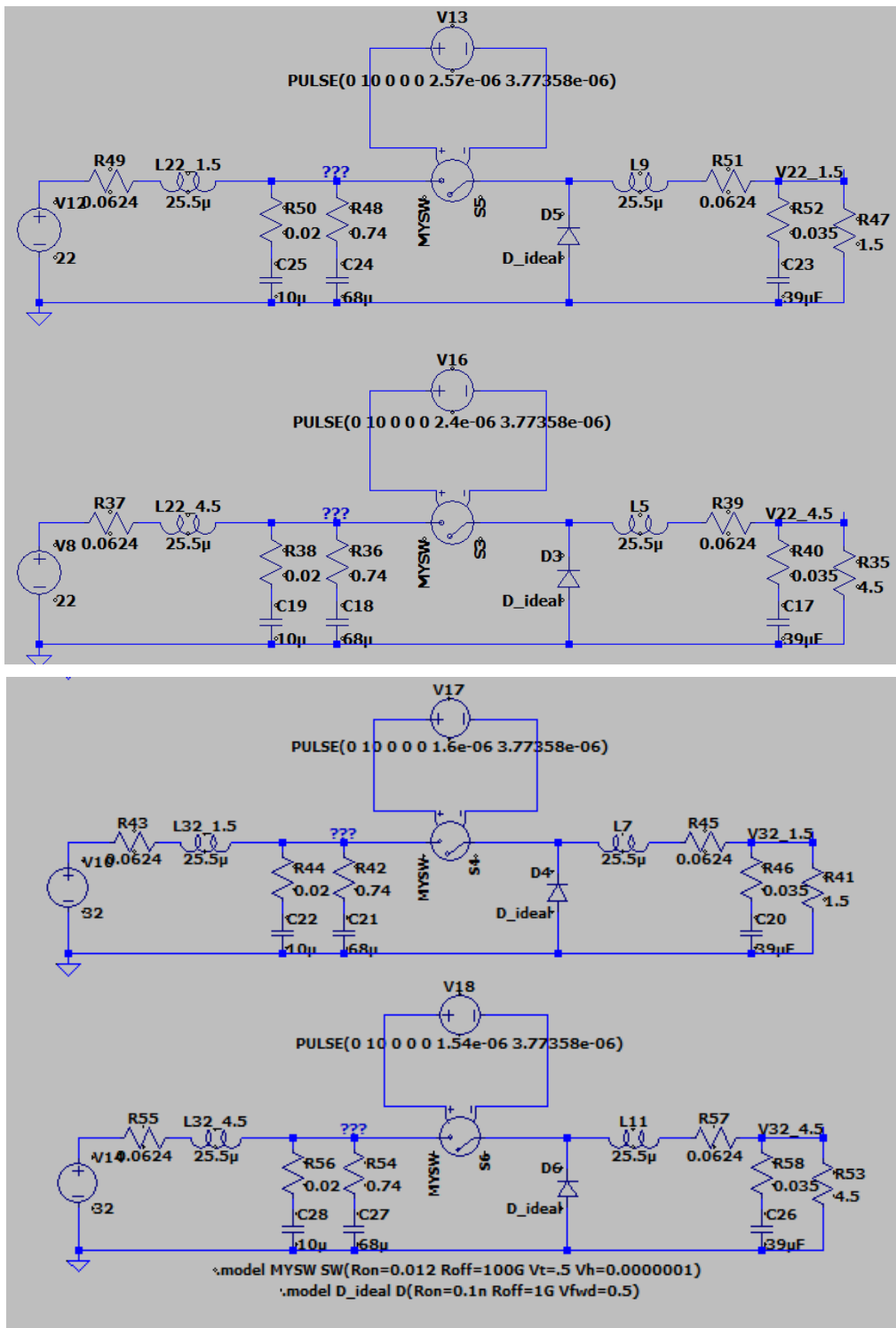
T_Lin = (P_AC_core_in + P_core_in + P_DC_core_in) * 23 + 50;

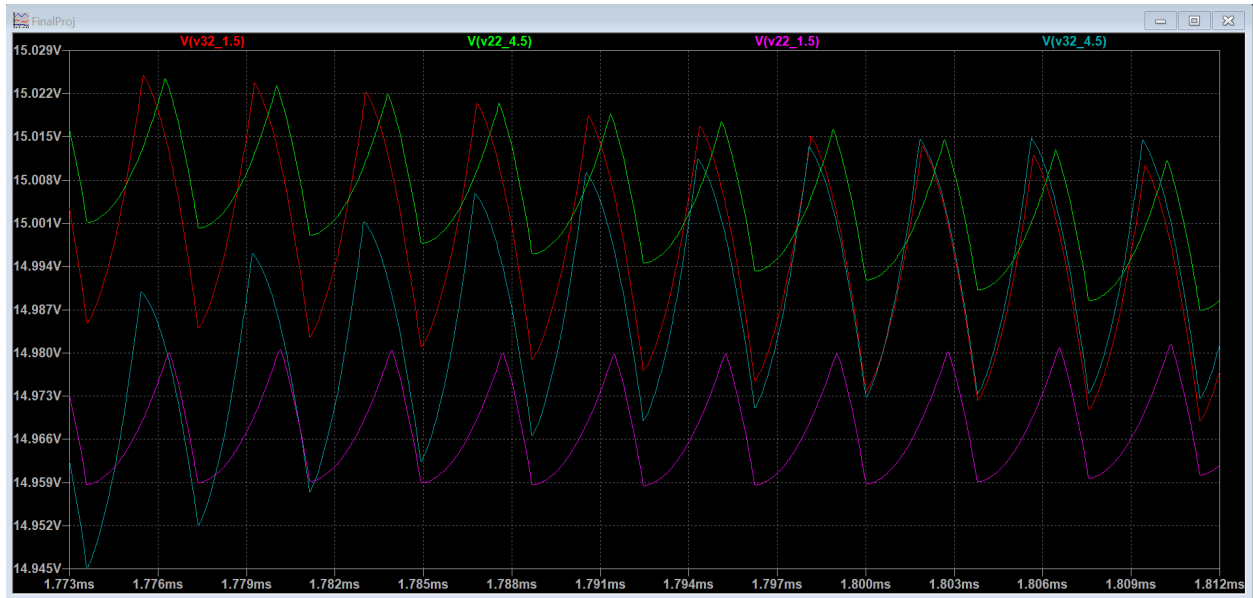
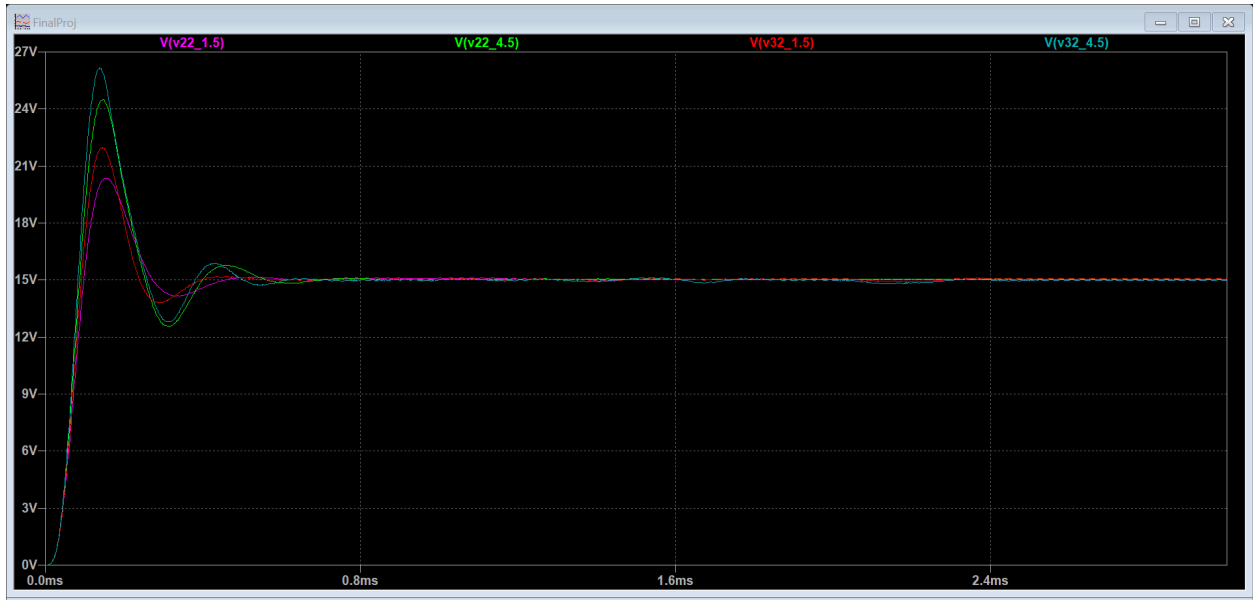
% Output Inductor

T_Lout = (P_AC_core_out + P_core_out + P_DC_core_out) * 23 + 50;

%%%

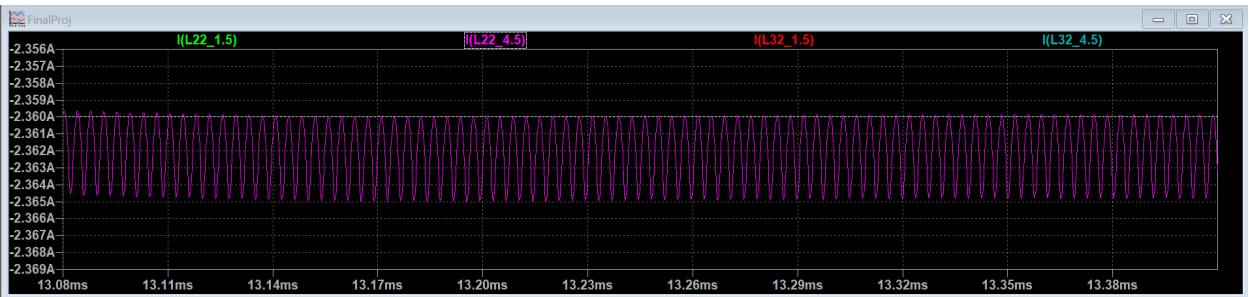
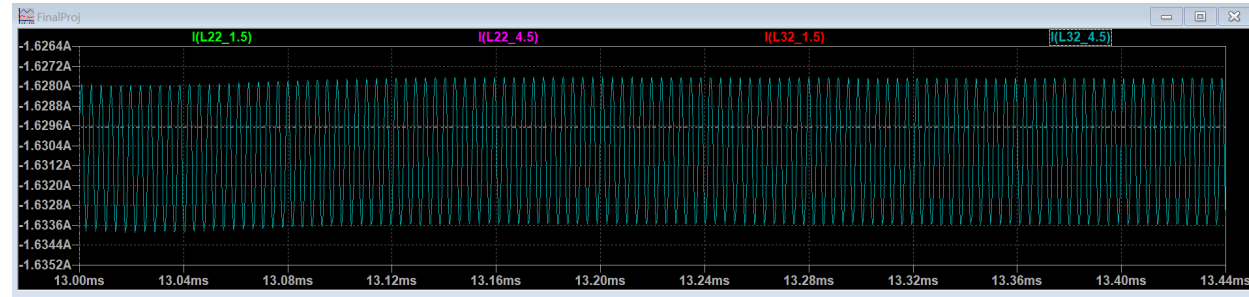
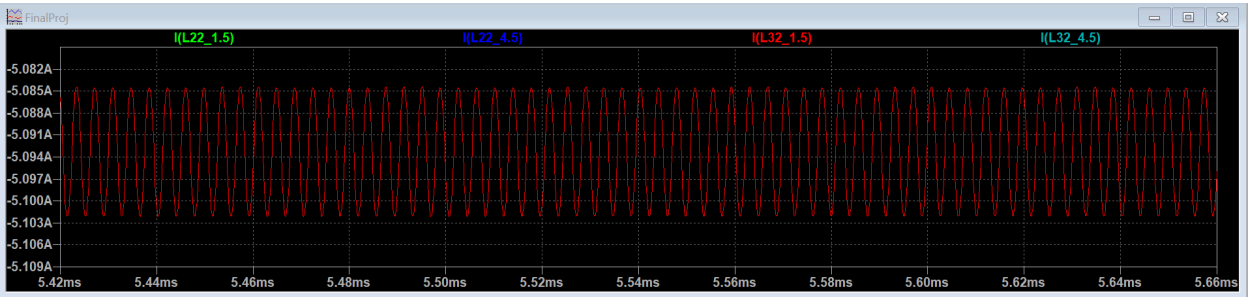
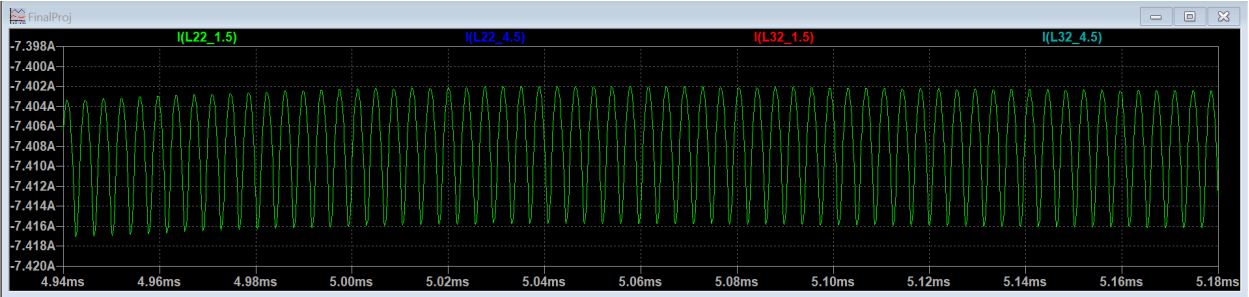
Open Loop Ripple Performance - SPICE





Open Loop voltage simulation - 4 boundary cases
(Overshoot is not a concern for this simulation, it's addressed in the closed loop control section. Main interest here is ripple)

Open Loop current simulation (ripples) - 4 boundary cases



Simulation Results:

- 22V input, 150W
 - Output Voltage Ripple, peak-peak ~ 20 mV
 - Input Current Ripple, peak-peak ~ 14 mA
- 22V input, 50W
 - Output Voltage Ripple, peak-peak ~ 24 mV
 - Input Current Ripple, peak-peak ~ 5 mA

- 32V input, 150W
 - Output Voltage Ripple, peak-peak ~ 37 mV
 - Input Current Ripple, peak-peak ~ 17 mA
- 32V input, 50W
 - Output Voltage Ripple, peak-peak ~ 45 mV
 - Input Current Ripple, peak-peak ~ 5.6 mA

All Values are within specs (Output Voltage: 80 mV_{pp}, Input current: 50 mA_{pp}). Not an exact match with the theoretical prediction (SPICE model does have more details about the devices, such as diode drop / parasitics / etc., when obtaining ripples. There might also be calculation error)

Feedback Controller deals with local average dynamics, so ripple performance in the feedback section should be identical to the ones as shown in this section.

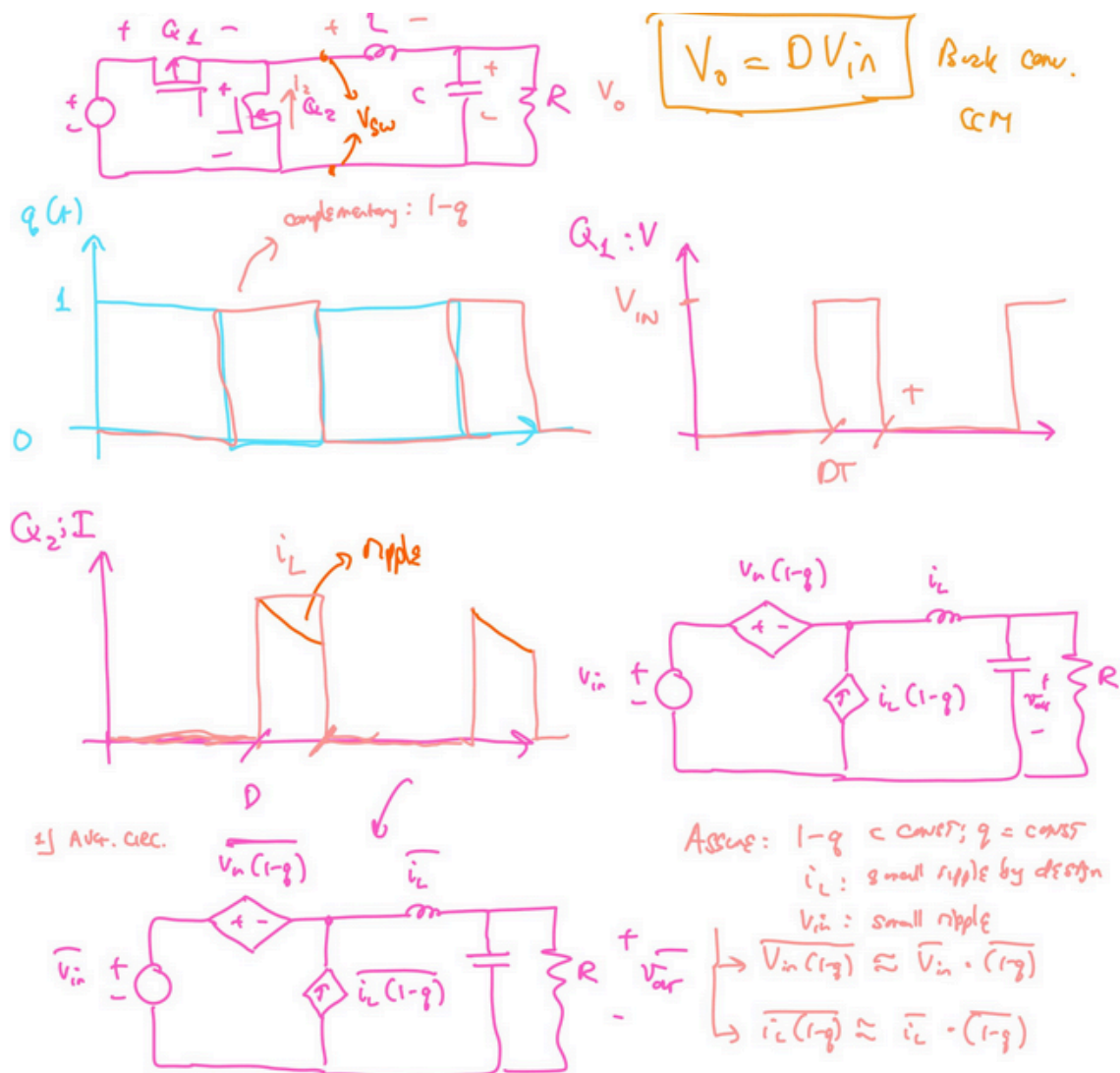
%%%

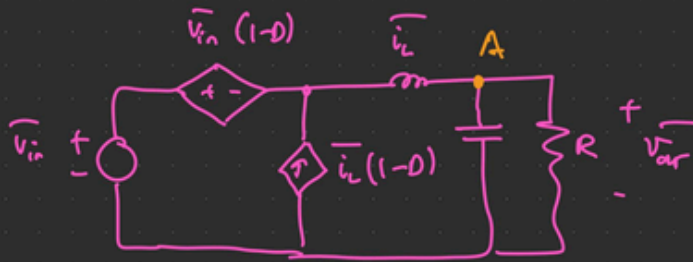
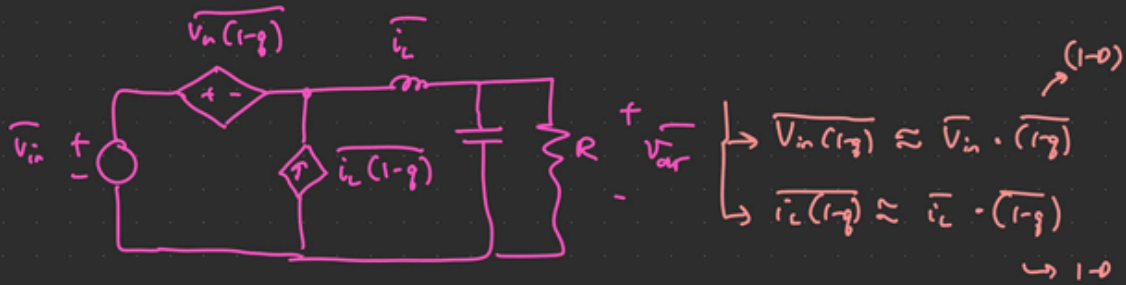
Controller Design

Buck Converter transfer function from duty cycle -> output can be represented with the following transfer function. Derivation follows.

$$Plant(s)_{D \text{ to } V_{out}} = \frac{V_{in}}{L_o \cdot C_o \cdot s^2 + \frac{L_o}{R_o} \cdot s + 1}$$

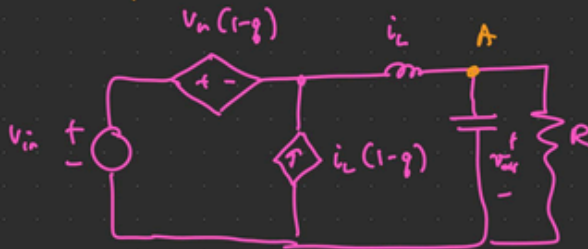
$$Plant(s)_{V_{in} \text{ to } V_{out}} = \frac{D}{L_o \cdot C_o \cdot s^2 + \frac{L_o}{R_o} \cdot s + 1}$$





$$\begin{aligned}
 & \bar{V}_{in} - \bar{V}_{in}(1-D) - \bar{V}_{out} = L \dot{\bar{i}}_L \quad \text{inductor current} \\
 & \hookrightarrow + \bar{V}_{in} D - \bar{V}_{out} = L \dot{\bar{i}}_L \\
 & - \frac{\bar{V}_o}{R} + \bar{i}_L = C \dot{\bar{V}}_o \quad \text{KCL node A}
 \end{aligned}
 \quad \left. \vphantom{\begin{aligned} & \bar{V}_{in} - \bar{V}_{in}(1-D) - \bar{V}_{out} = L \dot{\bar{i}}_L \\ & - \frac{\bar{V}_o}{R} + \bar{i}_L = C \dot{\bar{V}}_o \end{aligned}} \right\} \text{Average circuit}$$

State space:



State eqns:

$$\begin{aligned}
 & \bar{V}_{in} - \bar{V}_{in}(1-q) - \bar{V}_{out} = L \dot{\bar{i}}_L \\
 & \bar{i}_L - \frac{\bar{V}_{out}}{R} = C \dot{\bar{V}}_{out}
 \end{aligned}
 \quad \left. \vphantom{\begin{aligned} & \bar{V}_{in} - \bar{V}_{in}(1-q) - \bar{V}_{out} = L \dot{\bar{i}}_L \\ & \bar{i}_L - \frac{\bar{V}_{out}}{R} = C \dot{\bar{V}}_{out} \end{aligned}} \right\} \begin{array}{l} \text{inductor} \\ \text{node A KCL} \end{array}$$

Verification

$$\bar{V}_{in} D - \bar{V}_o = L \dot{\bar{i}}_L$$

Set $\dot{\bar{i}}_L = 0$

$$\rightarrow \bar{V}_o = D \bar{V}_{in}$$

boost converter eqn!

State
space

$$\begin{aligned}
 & \rightarrow \left\{ \begin{array}{l} \bar{V}_{in} - \bar{V}_{in}(1-q) - \bar{V}_o = L \dot{\bar{i}}_L \\ \bar{i}_L - \frac{\bar{V}_o}{R} = C \dot{\bar{V}}_o \end{array} \right. \\
 & \quad \quad \quad \approx \bar{V}_{in} \bar{q} = \bar{V}_o \\
 & \rightarrow \left\{ \begin{array}{l} + \bar{V}_{in} \bar{q} - \bar{V}_o = L \dot{\bar{i}}_L \\ \bar{i}_L - \frac{\bar{V}_o}{R} = C \dot{\bar{V}}_o \end{array} \right.
 \end{aligned}$$

Equivalent!
inductor



$$\begin{cases} + \bar{V}_{in} d - \bar{V}_{out} = L \dot{\bar{i}}_L \\ - \frac{\bar{v}_o}{R} + \bar{i}_L = C \dot{\bar{v}}_o \end{cases}$$

$$\begin{cases} \bar{v}_{in} d - \bar{v}_o = L \dot{\bar{i}}_L & \text{NOT linear! } \bar{v}_{in} d \text{ term} \\ - \frac{\bar{v}_o}{R} + \bar{i}_L = C \dot{\bar{v}}_o \end{cases}$$

$$\bar{v}_{in} = V_{in} + \tilde{v}_{in}$$

$$d = D + \tilde{d}$$

$$\bar{v}_o = V_o + \tilde{v}_o$$

$$\rightarrow \begin{cases} (V_{in} + \tilde{v}_{in})(D + \tilde{d}) - (V_o + \tilde{v}_o) = L(\dot{I}_L + \dot{\tilde{i}}_L) \\ - \frac{(V_o + \tilde{v}_o)}{R} + (I_L + \tilde{i}_L) = C(\dot{V}_o + \dot{\tilde{v}}_o) \end{cases}$$

$$\rightarrow \begin{cases} \cancel{V_{in} \cdot D} + \cancel{V_{in} \tilde{d}} + \cancel{\tilde{v}_{in} D} + \cancel{\tilde{v}_{in} \tilde{d}} - \cancel{V_o} - \tilde{v}_o = L(\cancel{\dot{I}_L} + \dot{\tilde{i}}_L) \\ - \cancel{\frac{V_o}{R}} - \frac{\tilde{v}_o}{R} + \cancel{I_L} + \tilde{i}_L = C(\cancel{\dot{V}_o} + \dot{\tilde{v}}_o) \end{cases}$$

S.S current drawn
by load = supplied
inductor current

Capital - only terms (should)
be gone!

$$\begin{cases} V_{in} \tilde{d} + \tilde{v}_{in} D - \tilde{v}_o = L \dot{\tilde{i}}_L \\ - \frac{\tilde{v}_o}{R} + \tilde{i}_L = C \dot{\tilde{v}}_o \end{cases} \rightarrow \begin{cases} V_{in} \tilde{d} + \tilde{v}_{in} D - \tilde{v}_o = L \dot{\tilde{i}}_L \\ - \frac{\tilde{v}_o}{R} + \frac{\tilde{i}_L}{C} = \dot{\tilde{v}}_o \end{cases}$$

$$\rightarrow \begin{cases} V_{in} \tilde{d} + \tilde{v}_{in} D - \tilde{v}_o = L \dot{\tilde{i}}_L \\ -\frac{\tilde{v}_o}{RC} + \frac{\tilde{i}_L}{C} = \dot{\tilde{v}}_o \end{cases}$$

To derive separate transfer function of one variable to another

for $\tilde{d} \rightarrow \tilde{v}_o$: first; set $\tilde{v}_{in} = 0$

$$\rightarrow \begin{cases} V_{in} \tilde{d} - \tilde{v}_o = L \dot{\tilde{i}}_L \\ -\frac{\tilde{v}_o}{RC} + \frac{\tilde{i}_L}{C} = \dot{\tilde{v}}_o \end{cases} \rightarrow V_{in} \tilde{d} - \tilde{v}_o = L \left(\dot{\tilde{v}}_o \cdot C + \frac{\tilde{v}_o}{R} \right)$$

thought: this is why we linearize & decouple terms!

$$\downarrow$$

$$\tilde{i}_L = \dot{\tilde{v}}_o \cdot C + \frac{\tilde{v}_o}{R}$$

$$V_{in} \tilde{d} = LC \ddot{\tilde{v}}_o + \frac{L}{R} \dot{\tilde{v}}_o + \tilde{v}_o$$

$\downarrow \mathcal{L}(s)$

$$V_{in} d(s) = v_o(s) \left[LCs^2 + \frac{L}{R}s + 1 \right]$$

$$\rightarrow \frac{v_o(s)}{d(s)} = \frac{V_{in}}{\left[LCs^2 + \frac{L}{R}s + 1 \right]}$$

So that we can consider them independently.

Though; only the parameters ($\tilde{d}$; $\tilde{v}_{in}$) should be set to zero! State variable remain as is!

Audio susceptibility: for $\tilde{v}_{in} \rightarrow \tilde{v}_o$. Set $\tilde{d} = 0$

$$\rightarrow \begin{cases} \cancel{V_{in} \tilde{d}} + \tilde{v}_{in} D - \tilde{v}_o = L \dot{\tilde{i}}_L \\ -\frac{\tilde{v}_o}{RC} + \frac{\tilde{i}_L}{C} = \dot{\tilde{v}}_o \end{cases}$$

$$\downarrow$$

$$\tilde{i}_L = \dot{\tilde{v}}_o \cdot C + \frac{\tilde{v}_o}{R}$$

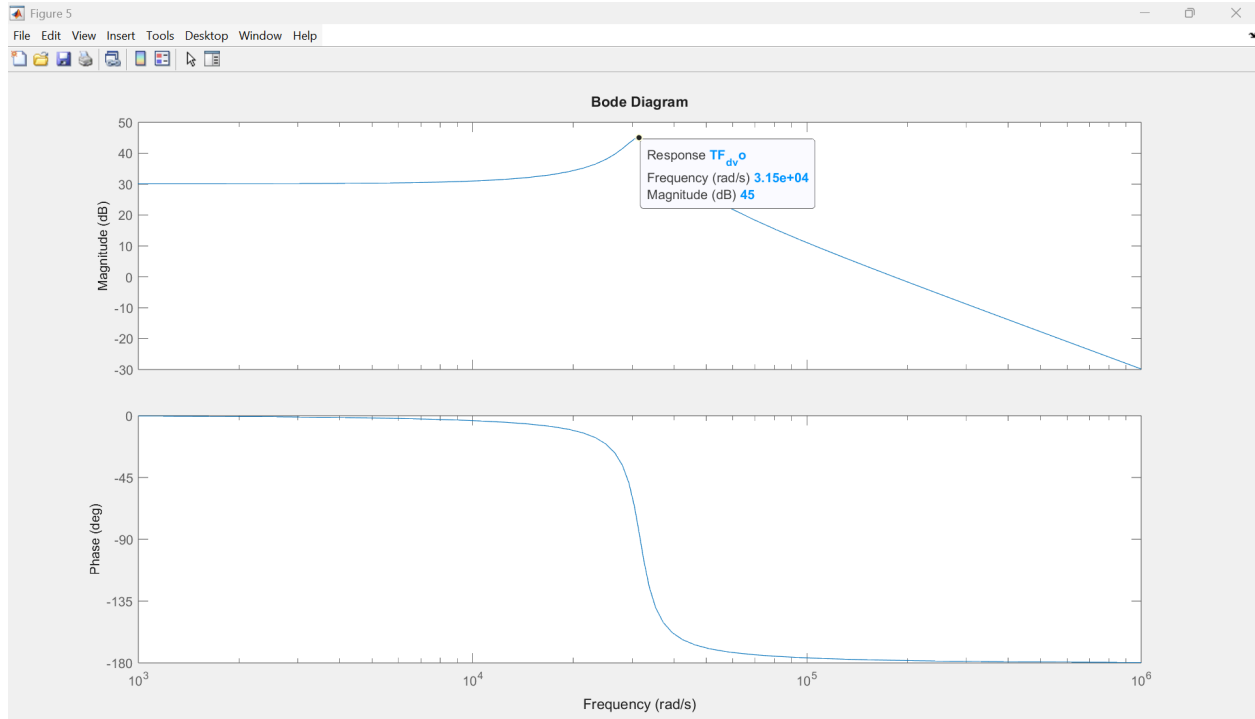
$$V_{in} \tilde{d} = LC \ddot{\tilde{v}}_o + \frac{L}{R} \dot{\tilde{v}}_o + \tilde{v}_o$$

$\downarrow \mathcal{L}(s)$

$$v_{in}(s) D = v_o(s) \left[LCs^2 + \frac{L}{R}s + 1 \right]$$

$$\rightarrow \frac{v_o(s)}{v_{in}(s)} = \frac{D}{\left[LCs^2 + \frac{L}{R}s + 1 \right]}$$

Transfer Function Derivations



Example Plant Bode Plot (D→Vout). Notice the resonant behavior

Controller Design Strategy:

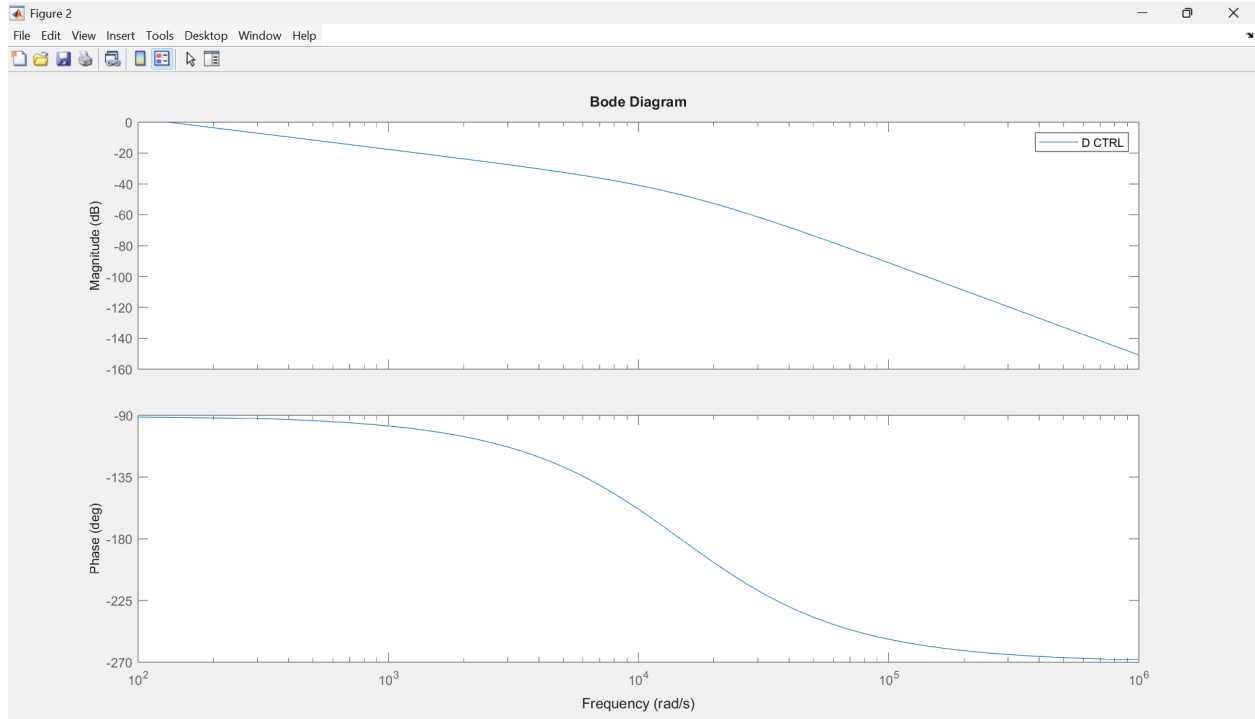
- Frequency Control Method (Goal is to achieve at least 45 degree of phase margin for 4 boundary operating conditions).
- Plant in question: 2nd order system with resonant frequency
 - The resonant behavior creates complication for designing crossover frequency
 - Even if the resonant does not crossover, oscillation will be seen if this resonant is not attenuated appropriately
 - One approach: Crossover can be done by using an integral (lag) controller and design the crossover frequency to be before the resonant frequency
 - Integral introduces a “slope” in magnitude, allowing crossover to happen. It also introduces a -90 degree of phase lag.
 - Very low crossover frequency acts like 1st order frequency.
- *I choose this approach.*

- One can attempt to try crossing over after the resonant frequency
 - Lead Controller is crucial to introduce more phase margin after the crossover frequency (there will be none)
 - Very fast settling time, but require the driver to be able to output large/fast command signal
- If choosing a PI controller for this plant, then resonant must be carefully taken care of:
 - LPF (two of them) are introduced to filter out the resonant frequency.
 - LPF's cut-off frequency should be sufficiently higher than crossover frequency so as to not introduce degradation in phase margin.
- Open Loop Transfer Function (Plant x Controller) is shown in the performance report. Very important plot!
- In addition discussion:
 - DC gain affects the steady state settling tolerance. Higher is better.
 - Integral Control theoretically has infinite open-loop DC gain.
 - Having ample gain margin is also a good thing.
 - Ample phase and gain margin help the design to be resilient against variation / inaccuracy in modelling.

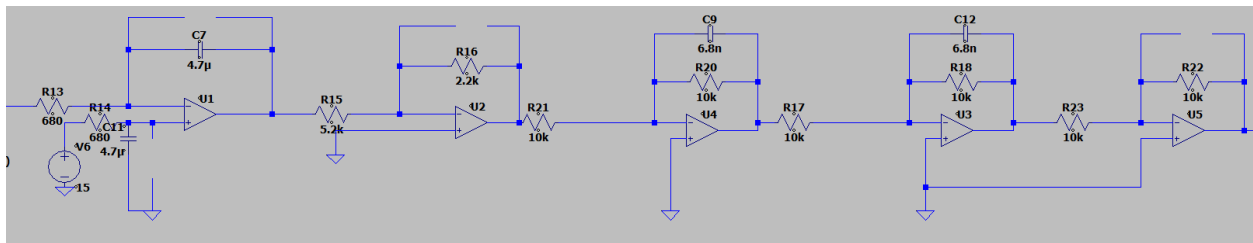
```

29 TF_d_vo = Vin / (L*Co*s^2 + L/R*s + 1);
30 TF_vi_vo = D / (L*Co*s^2 + L/R*s + 1);
31 %% Controller Design
32 s = tf('s');
33 % Kp
34 K_p = 2200/5200; %setting the crossover
35 %Integral_
36 w_lag = (4.7*10^(-6))*680;
37 Integral = 1/(w_lag*s);
38
39 %LPF 1
40 LPF1 = (10000/10000) * 1/(10000*6.8*10^(-9)*s + 1);
41
42 %LPF2 - identical to LPF 1
43 LPF2 = (10000/10000) * 1/(10000*6.8*10^(-9)*s + 1);
44
45 % Controller Creation
46 CTRL = K_p * Integral * LPF1 * LPF2 ;
47 % figure()
48 % bode(CTRL);
49

```



Controller Transfer Function and Plot: Proportional, Integral and two LPF. Crossover is designed around the 2×10^3 rad/s region (where there's “ample” amount of phase margin).



Controller Implementation in Circuit/SPICE using Op-Amp in inverting configuration. An extra unity inversion is needed at the end.

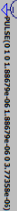
```

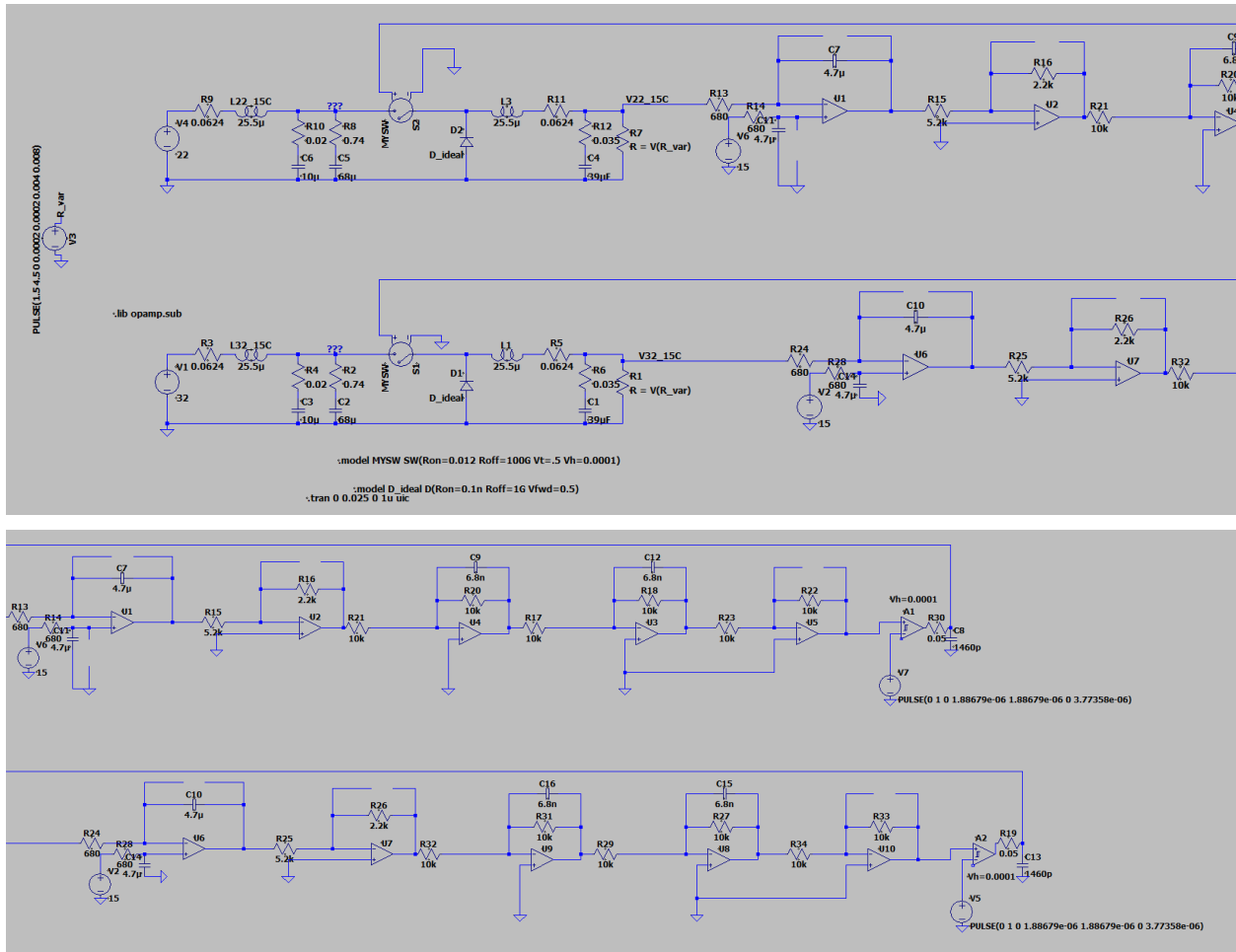
51 %% Margin Calculation
52 FW_d_vo = CTRL * TF_d_vo;
53 %FW_vi_vo = CTRL * TF_vi_vo;
54
55 figure()
56 margin(FW_d_vo)
57 % hold on
58 % bodeTF_d_vo)
59 hold on
60 bode(CTRL)
61 legend("D CTRL", "CTRL")
62
63
64 close_loop = FW_d_vo/(1+FW_d_vo);
65 figure()
66 bode(close_loop)
67 figure()
68 step(close_loop)
69
70 %%
71 figure()
72 bode(TF_d_vo)

```

**Code to generate gain and phase Margin of Open Loop TF
Step response is also generated to validate normalized overshoot.
Refer to the Performance Report Section for results.**

Controller and Closed Loop Transient Performance - SPICE



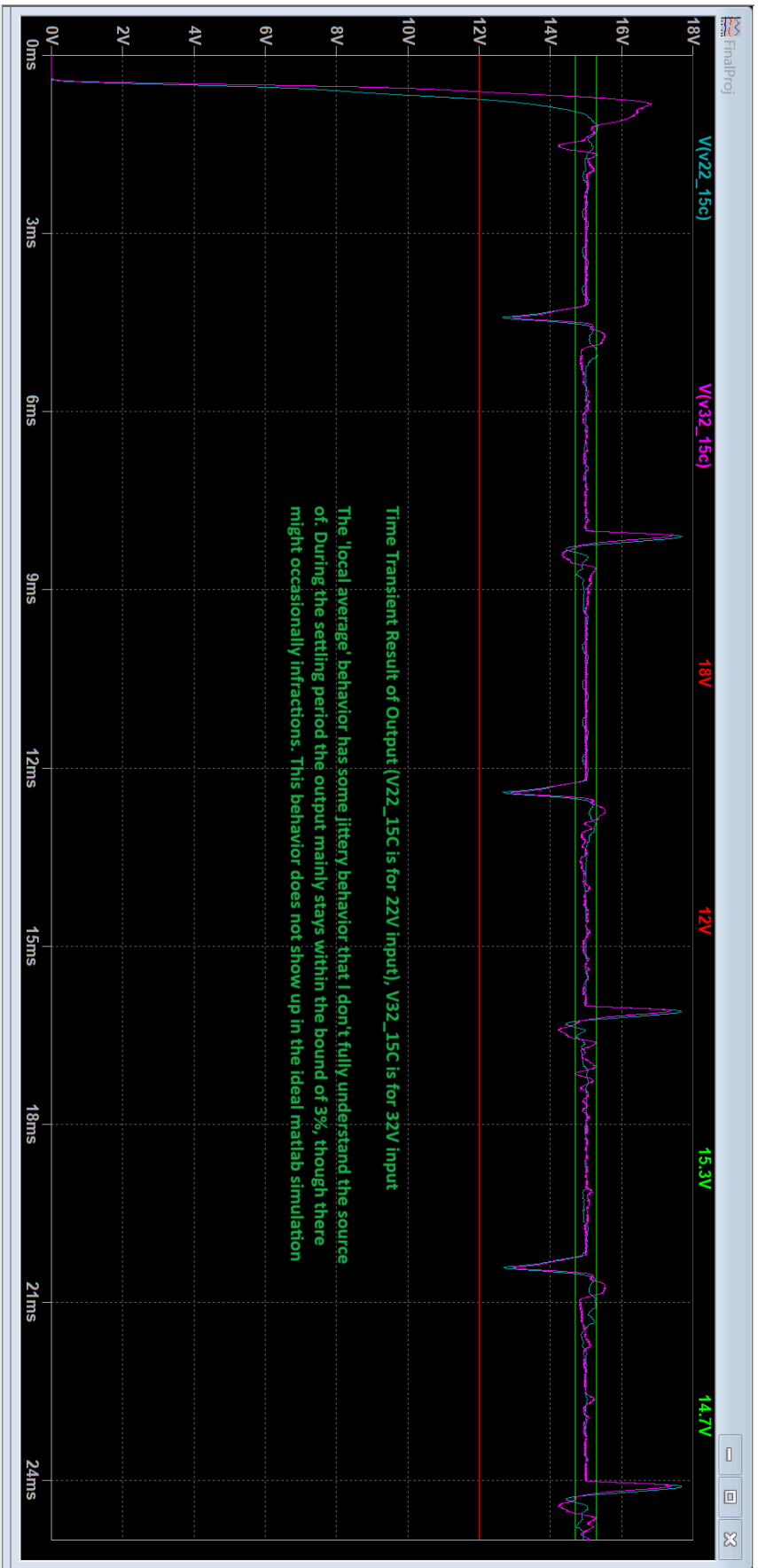


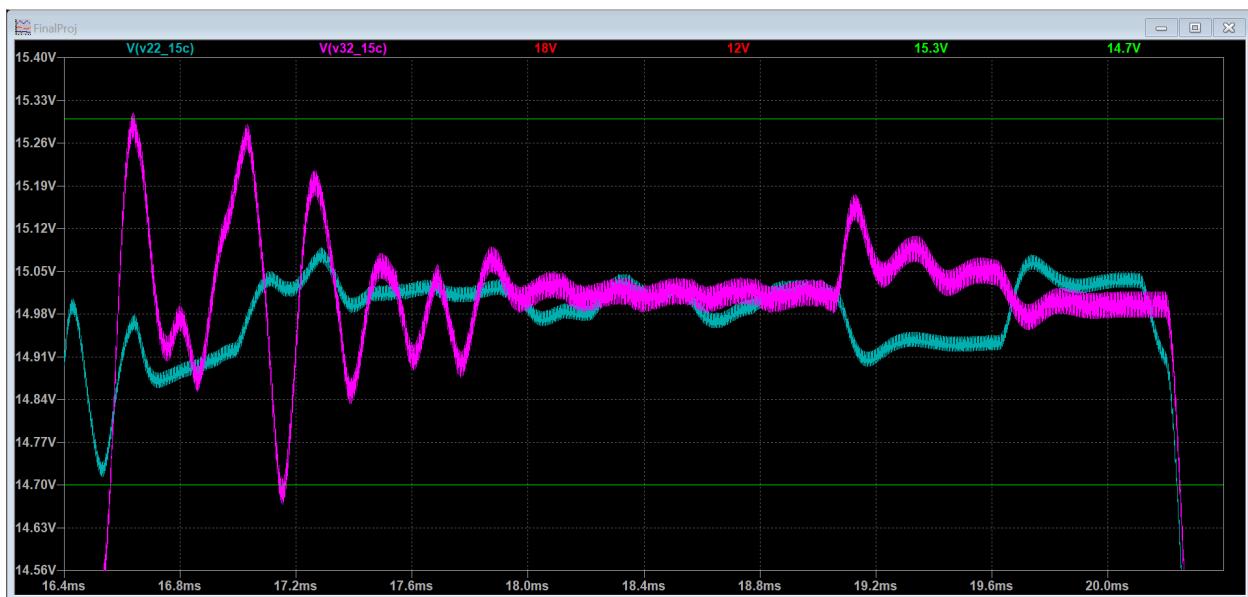
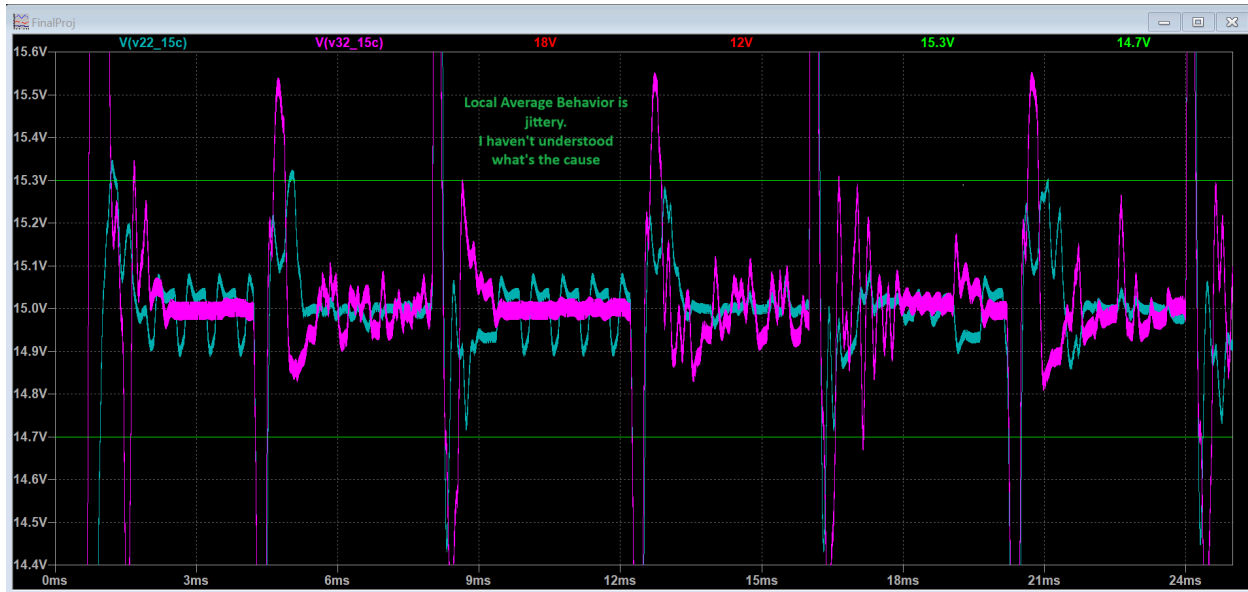
Controller Design: PI Controller with 2 1st-order LPF.

Also employ Triangle-Modulation PWM Scheme (implemented with a comparator at the end of the signal chain) to generate PWM

Capacitor is added after the comparator as a simple attempt to simulate FET input capacitance and driver dynamics. Also assume the driver has output impedance of 0.05 Ohm and can source sufficient current to charge the gate capacitor. (Input Capacitance is 1460 pF - a more in-depth design would take the driver into account and making sure it can drive this load)

Resistive Load is switched using the voltage source on the left side of the SPICE model)

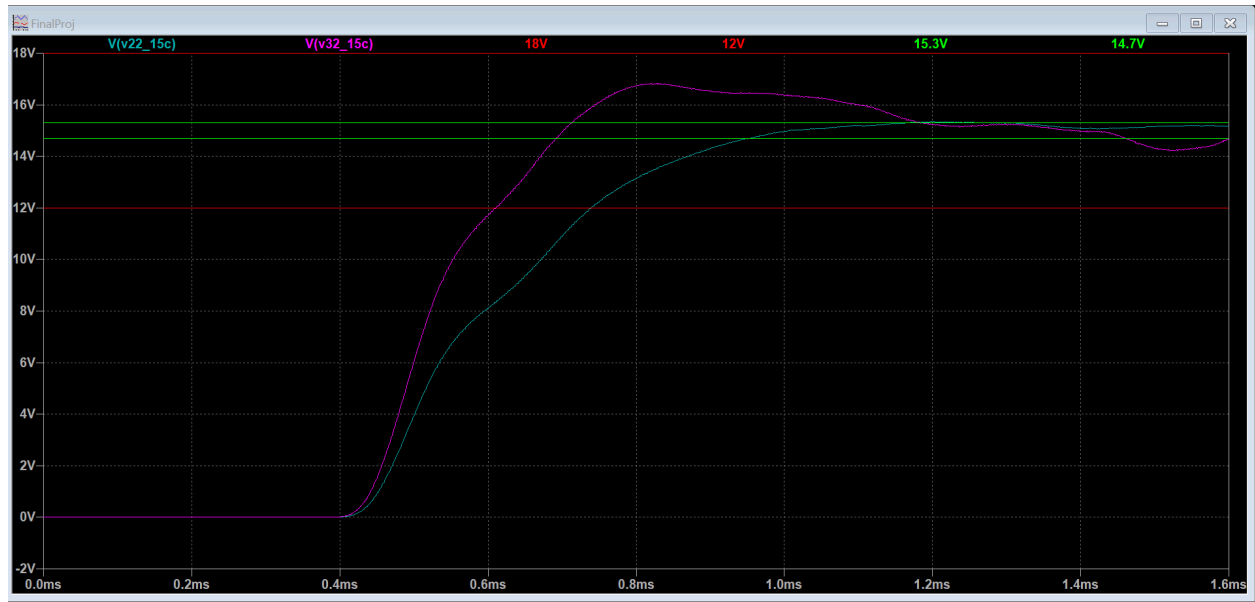




The signal does not overshoot the 20% limit (12V ~ 18V) during transient jump

I haven't fully understood the erratic "local average" behavior observed in the waveform, though it does not exceed the 3% threshold most of the time. I'm not sure what the reason for this erratic behavior is. There's occasional infraction happening as shown in the above image where the waveform hit the 3% limit.

The voltage values "hovers" around 15V.



Step response behavior resembles the performance report section!

%%%

Thermal Design for Semiconductor Components (FET & Diode)

The number (referring to the performance report) shows that for the worst case scenario (for a single component), FET experiences up to 3.13W of total power. Thus the thermal design incorporating additional margin is designed to sustain 4W or more.

As the exposed heat pads of both components are facing toward the board (likely soldered), heat dissipation process needs to be done through the board and to the other side. Thus the PCB board should incorporate appropriate amount of thermal vias

Derived Information from AN-2020 document relevant information:

θ_{VIA}	261 °C / W	Thermal resistance of typical 12 mil via.	Via length = 0.165 cm (65 mils), 0.5 ounce copper plating thickness = 0.00175 cm, drill hole radius = 6 mil (0.01524 cm), thermal conductivity of copper (λ_{Cu}) = 4 W / (cm °C)	$\theta_{VIA} = \frac{\frac{1}{\lambda_{Cu}} \times \text{Length}}{\text{Area}}$ $= \frac{0.25 \frac{^{\circ}\text{C cm}}{\text{W}} \times 0.165 \text{ cm}}{\pi \times [(0.01524 \text{ cm})^2 - (0.01524 \text{ cm} - 0.00175 \text{ cm})^2]}$ (3)
----------------	------------	---	---	---

θ_{FR4}	13.9 °C / W	Vertical thermal resistance of FR-4 substrate.	1 cm square, FR-4 thickness = 0.032 cm (12.6 mil), thermal conductivity of FR-4 (λ_{FR4}) = 0.0023 W / (cm °C)	$\theta_{FR4} = \frac{\frac{1}{\lambda_{FR4}} \times \text{Thickness}}{\text{Area}} = \frac{435 \frac{^{\circ}\text{C cm}}{\text{W}} \times 0.032 \text{ cm}}{1 \text{ cm} \times 1 \text{ cm}}$ (5)
----------------	-------------	--	--	---

- Assuming 62 mil thick board, and 12 mil via (0.3mm diameter)
 - FR4: 13.9x5 = 69.5 C/W
 - Via: 261 C/W per individual via

Thermal Pad and Heat Sink is mounted on the other side of the board, and tasked with dissipating the heat. Area of the thermal pad for each design is set to be similar to the solder pad, which is a conservative estimation.

Actual design can have a larger thermal pad to capture more heat path as heat also spread “radially” and not just perfectly vertical through the PCB

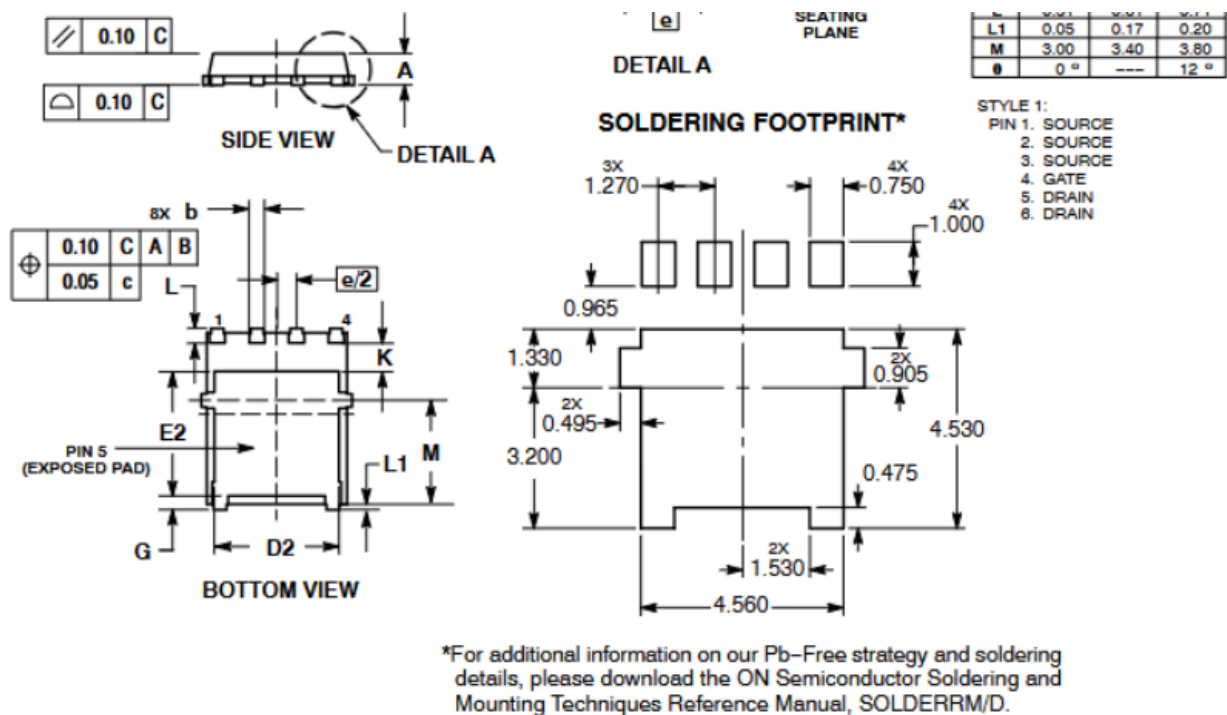
FET and Diode heat dissipation solutions are detailed as follow:

FET

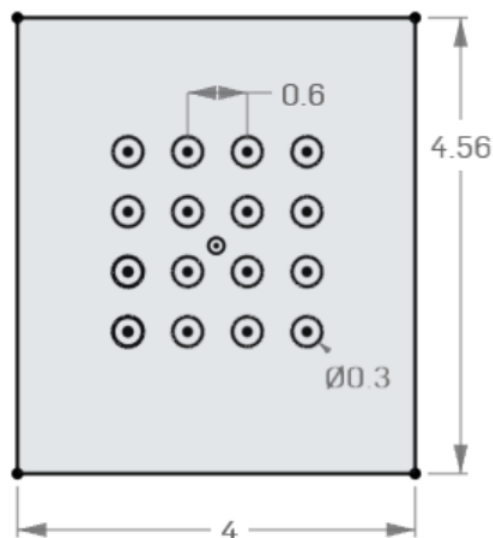
Worst Case Dissipation: 3.13W. Design for 4W

THERMAL RESISTANCE MAXIMUM RATINGS

Parameter	Symbol	Value	Unit
Junction-to-Mounting Board (top) – Steady State (Notes 2, 3)	$R_{\Psi U-mb}$	1.4	°C/W
Junction-to-Ambient – Steady State (Note 3)	$R_{\theta JA}$	41	



- Thermal Impedance Junction to Solder Point: 1.4 C/W
- Pad Area: ~ 4.560 mm x 4.0mm. 16 12-mil vias can fit inside.
 - Effective via stacks thermal impedance is $R_{via_stack} = 261/16 = 16.3$ C/W
- Effective Solder Point to Thermal Pad through FR4 and viastack is thus: $(R_{FR4} // R_{via_stack}) = 69.5 // 21.75 = 13.2$ C/W

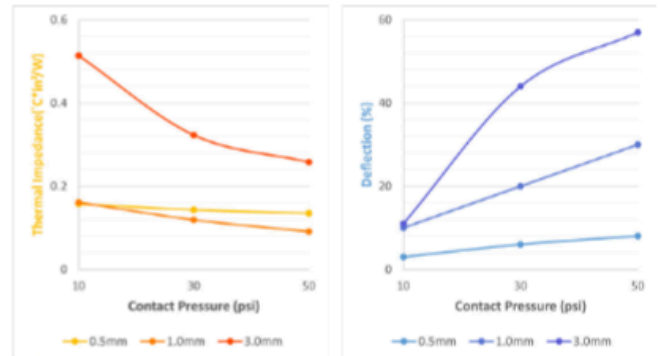


-
- **Thermal Pad:**
 - Thermal Conductivity: 16.8 W / K *m (lower end), Area is 4.560mm x 4.0mm, thickness is 2 mm

Properties



Contact Pressure, Thermal Impedance, and Deflection



Properties	TG-A1780	Unit	Tolerance	Test Method
Thermal Conductivity	17.8	W/m•K	±10%	ASTM D5470 Modified
Thickness	0.5~2.0	mm	-	ASTM D374
	0.0197~0.0787	inch	-	ASTM D374
Color	Light Gray	-	-	Colorimeter CIE 1976

- **Thermal Impedance of Thermal Pad:** = thickness / (conductivity * Area) = 6.654 C / W
- **Heat Sink:** For Natural Convection: Thermal Impedance ~6 C / W is a good estimation based on the table below (HSB28-606022). Note that the heat sink must be pressed against the heat pad with sufficient pressure (>50PSI)

Power [W]	Heatsink Temperature Rise Above Ambient (ΔT = T _{hs} - T _a) [°C]		
	Natural Conv.	200 LFM	400 LFM
0	0	0	0
1	5.1	1.4	0.9
3	17.9	4.4	2.7
5	29.3	7.6	4.6
7	38.8	10.9	6.5
9	48.0	14.2	8.4
11	56.5	17.5	10.4
13	64.4	20.7	12.2
15	72.1	23.9	14.1
17	79.7	27.2	16.0
19	86.8	30.4	18.0
21	93.9	33.8	19.9
23	100.9	37.2	21.8
25	107.7	40.6	23.8

T_{hs}: "hot spot" temperature measured on the heatsink
T_a: ambient temperature

- Tally and Check:
 - Junction to Solder Pad: 1.4 C/W
 - FR4 and Via Stack: 13.2 C/W
 - Thermal Pad: 6.654 C/W
 - Heat Sink: 6 C/W
 - Total Impedance (Junction to Ambient): 27.254 C/W
 - For 4W of power: Junction Temperature @ 40C ambient:
 - $27.254 * 4 + 40 = 149C < 175 C$
 - Reminder that 4W is conservative - worst calculated case is 3.13W
 - Limit of this design for 175 C Junction Temperature: 4.95 W

THE VALUE OF 27.254 C/W IS USED FOR FET TEMPERATURE CALCULATION
THAT'S GENERATED IN THE PERFORMANCE REPORT

Diode

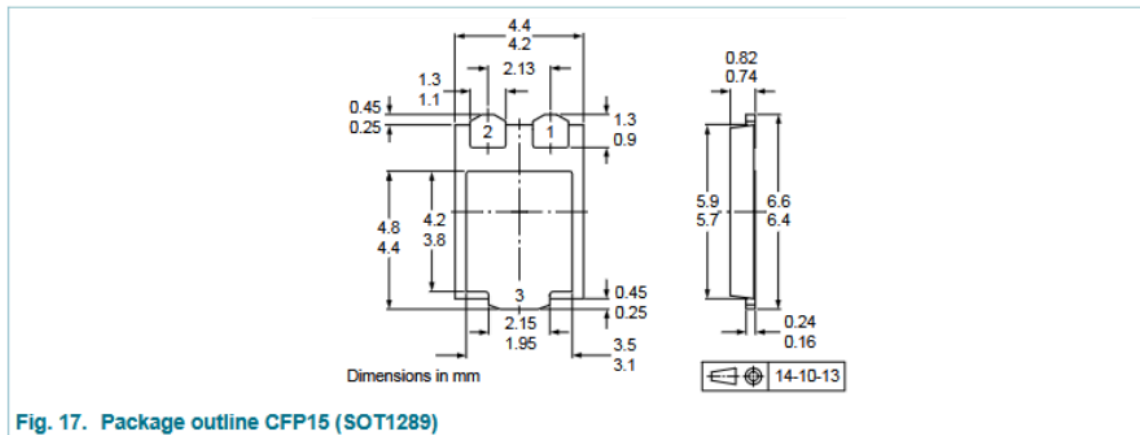
Worst Case Dissipation: 2.65W. Design for 4W

Table 6. Thermal characteristics

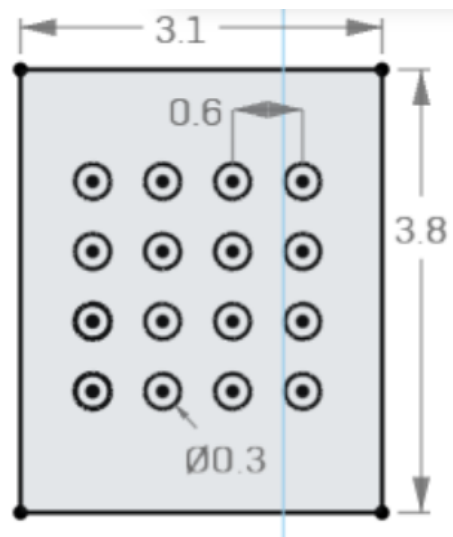
Symbol	Parameter	Conditions		Min	Typ	Max	Unit
$R_{th(j-a)}$	thermal resistance from junction to ambient	in free air	[1][2]	-	-	90	K/W
			[1][3]	-	-	70	K/W
			[1][4]	-	-	40	K/W
$R_{th(j-sp)}$	thermal resistance from junction to solder point		[5]	-	-	3	K/W

- Thermal Impedance from Junction to Solder Point: 3C/W

12. Package outline



- Thermal Impedance Junction to Case: 3 C/W
- Cross Sectional Area of the Pad: Conservative: 3.1 mm x 3.8 mm
- Same via (16 12-mil vias) design and board thickness as FET:
 - Reuse result: Thermal Impedance from Solder Point to Thermal Pad through FR4 and via-stack is thus: $(R_{FR4} \parallel R_{via_stack}) = 69.5 \parallel 16.3 = 13.2 \text{ C/W}$



-
- Thermal Pad: Use similar design, but with a different area:
 - Thermal Conductivity: 16.8 W / K *m (lower end), Area is 3.1mm x 3.8mm, thickness is 2 mm
 - Thermal Impedance of Thermal Pad: = thickness / (conductivity * Area) = 10.11 C/W
- Use the same heat sink design as FET (HSB28-606022).: 6 C/W
- Tally:
 - Junction to Solder Point: 3 C/W
 - FR4 and Via Stack: 13.2 C/W
 - Thermal Pad: 10.11 C/W
 - Heat Sink: 6/W
 - Total Impedance (Junction to Ambient): 32.31 C/W
 - For 4W of power: Junction Temperature @ 40C ambient:
 - $32.31 * 4 + 40 = 169.24C < 175 C$ (close!)
 - But this will definitely leave enough margin during normal operation
 - 4W is conservative - worst calculated case is 2.65W
 - Limit of this design for 175 C Junction Temperature: 4.18W

THE VALUE OF 27.254 C/W IS USED FOR DIODE TEMPERATURE CALCULATION THAT'S GENERATED IN THE PERFORMANCE REPORT

FET & Diode Temperature Calculation Routine

Temperature must not exceed 150C

```

231 % Diode
232 R_Diode_JA = 32.31;
233 T_diode = 40 + R_Diode_JA*P_diode %
234 % FET
235 R_FET_JA = 27.254;
236 T_FET = 40 + R_FET_JA*(P_cond_FET + P_sw_FET)

```

Appendix I: Design Code

```
%%
close all;
clear all;
clc;
function [B_core, B_AC_ripp, Inductance_method2, P_core_out, I_density_max, P_DC, P_AC, R_DC] =
inductor_metric(struct)
i_avg = struct.i_avg;
fsw = struct.fsw;
i_Lo_rip_pp = struct.i_Lo_rip_pp;
N_turns = struct.N_turns;
D_wire = struct.D_wire;
L_Airgap = struct.L_Airgap;
L_core_eff = struct.L_core_eff;
A_core_eff = struct.A_core_eff;
A_core_min = struct.A_core_min;
mu_rel = struct.mu_rel;
mu_o = struct.mu_o;
A_L = struct.A_L;
V_core = struct.V_core;
D_core = struct.D_core;
i_L_max = i_avg + i_Lo_rip_pp/2;
MMF = i_L_max * N_turns;
MMF_ripp = max(i_Lo_rip_pp/2) * N_turns;

Reluc_core = L_core_eff/(mu_o*mu_rel*A_core_eff);
Reluc_gap = L_Airgap/(mu_o*A_core_eff);
Reluc_eff = Reluc_core + Reluc_gap;

Flux = MMF/Reluc_eff;
Flux_AC = MMF_ripp / Reluc_eff;
Flux_Linkage = N_turns*Flux;
Inductance = N_turns^2/Reluc_eff;
Inductance_method2 = A_L*N_turns^2;
B_core = Flux/A_core_min;
B_AC_ripp = Flux_AC / A_core_min;
I_density_max = (i_avg+i_Lo_rip_pp/2)/(pi * D_wire^2 / 4);
C_mu = 2.5*10^(-4);
alpha_core = 1.63;
beta_core = 2.45;
P_core_out = C_mu * (fsw ^ alpha_core) * ((B_AC_ripp*1/2)^beta_core) * V_core ;
res_per_length = 3.277*10^(-3); % 10 AWG
R_DC = D_core*pi*res_per_length*N_turns;
delta_skin = 0.00013;
W_delta_ratio = D_wire/delta_skin;
R_AC = R_DC * W_delta_ratio * (sinh(2*W_delta_ratio) + sin(2*W_delta_ratio))/(cosh(2*W_delta_ratio) -
cos(2*W_delta_ratio))
fprintf("R_AC is" + R_AC + "\n")
P_DC = i_avg^2 * R_DC;
P_AC = (i_Lo_rip_pp/2)^2 * R_AC;
end
%% Operating Condition
Vin = 22;
Vout = 15;
PWR = 150;
% User Defined
fsw = 265*10^3; % 1.67 x 10^7 rad/s
Tsw = 1/fsw;
%% Converter Operation Requirements and Derived Requirement
Eff_target = 0.9;
lim_vppout_rip = 0.08;
```

```

lim_ippin_rip = 0.05;
lim_vout_static = Vout * 0.03;
lim_vout_trans = Vout * 0.2;
T_low = -30;
T_high = 40;
%% Derived Parameters
Dccm = Vout ./ Vin;
I_out = PWR ./ Vout;
R_conv_neg = -(Vin)^2/(PWR) %  $r = P / (i_{in}^2) = P / (P/vin)^2 = vin^2 / P$ . Lowest power highest Vin is worst case
I_in = PWR ./ Vin;
%% Derived Parameters - EMI Target
isource_fundamental = I_out./pi.*sqrt(2-2*cos(2*pi.*Dccm));
% Maximum Filter Characteristic Frequency. Smaller Value Require Larger L/C
EMI_TF_dB_attenuation_factor = mag2db(lim_ippin_rip*(1/2) / isource_fundamental);
win_target = sqrt(lim_ippin_rip*1/2 .* 1./(isource_fundamental) * (2*pi*fsw)^2);
win_target_conv_factor = sqrt(lim_ippin_rip*1/2 .* 1./(isource_fundamental));
%% Component Selection
% Output: Inductor, Capacitor
Co = 39 * 10^(-6);
Resr_Co = 0.035;
Lesr_Co = 1*10^(-9);
Lo = 25.5 * 10^(-6);
Resr_Lo = 0.0624
% Input: Inductor, Capacitor
Cin = 10 * 10^(-6);
Resr_Cin = 0.02;
Lesr_Cin = 1*10^(-9);
Lin = 25.5*10^(-6);
Resr_Lin = 0.0624; % change with inductor design!
% Input: Rd/Cd network
Cd = 68*10^(-6);
%Rd optimum in calculated below
% Input: Inductor, Capacitor, RdCd network
%% Output Ripple Derivation
% OUTPUT
i_Lo_rip_pp = Vout.*(1-Dccm).*Tsw./Lo;
% Reminder: for ripple model: Constant Current load is open! -> i ripple
% goes into C rippl = i = C dv dt -> integral (i) /C = dv. Inductor Current ripple
% has triangle shape.
v_Co_rip_pp = Vout.*(1-Dccm).*Tsw^2/(8*Lo*Co);
%% Input Filter FR
s=tf('s');
Z_Cin_real = 1/(Cin*s) + Resr_Cin + Lesr_Cin*s;
Z_Lin_real = Lin*s + Resr_Lin;
Zf_char = sqrt(Lin/Cin);
zeta_in = Cd / Cin;
f_f_input = 1/(2*pi*sqrt(Lin*Cin));
% formula to find an optimal Rd resistor value
Rd = Zf_char * sqrt((2+zeta_in)*(4+3*zeta_in)/((2*zeta_in^2)*(4+zeta_in)));
Z_d = 1/(Cd*s) + Rd;
H_s_ideal = (1 + Rd*Cd*s)/(1 + Rd*Cd*s + Lin*(Cd+Cin)*s^2 + Lin*Rd*Cd*Cin*s^3);
H_s_real = ((1/Z_d + 1/Z_Cin_real + 1/(Z_Lin_real))^(-1)) / (Z_Lin_real);
H_s_with_converter_load = ( (1/Z_d + 1/Z_Cin_real + 1/(Z_Lin_real) + 1/(R_conv_neg))^(-1) ) / (Z_Lin_real);
figure();
bode(H_s_ideal, "b");
hold on
bode(H_s_real, "r");
bode(H_s_with_converter_load)
legend("Ideal", "Real Component", "Real Component with Converter Load")
%% EMI Performance
i_input_rippl = isource_fundamental * f_f_input^2/fsw^2;

```

```

%% Inductor Output Evaluation
% 10 AWG
%RM
L_input.N_turns = 9;
L_input.fsw = fsw;
L_input.D_wire = 0.003264;
L_input.i_avg = I_in;
L_input.i_Lo_rip_pp = i_input_rippl*2 % i_Lo_rip_pp;
L_input.L_Airgap = 550*10^(-6);
L_input.L_core_eff = 0.0566;
L_input.A_core_eff = 0.000146;
L_input.A_core_min = 0.000125;
L_input.mu_rel = 97;
L_input.mu_o = 0.000001256637061;
L_input.A_L = 0.000000315;
L_input.V_core = 8340*0.001;
L_input.D_core = 0.025;
L_output.N_turns = 9;
L_output.fsw = fsw;
L_output.D_wire = 0.003264;
L_output.i_avg = I_out;
L_output.i_Lo_rip_pp = i_Lo_rip_pp % i_Lo_rip_pp;
L_output.L_Airgap = 550*10^(-6);
L_output.L_core_eff = 0.0566;
L_output.A_core_eff = 0.000146;
L_output.A_core_min = 0.000125;
L_output.mu_rel = 97;
L_output.mu_o = 0.000001256637061;
L_output.A_L = 0.000000315;
L_output.V_core = 8340*0.001;
L_output.D_core = 0.025;
[B_core_out, B_AC_rippl_out, Lout, P_core_out, Lout_current_density, P_DC_core_out,
P_AC_core_out, Resr_Lout] = inductor_metric(L_output);
[B_core_in, B_AC_rippl_in, Lin, P_core_in, Lin_current_density, P_DC_core_in, P_AC_core_in, Resr_Lin] =
inductor_metric(L_input);
%% Output Filter FR
s=tf('s');
Z_Cout_real = 1/(Co*s) + Resr_Co + Lesr_Co*s;
Z_Lout_real = Lo*s + Resr_Lo;
figure()
bode(Z_Cout_real / (Z_Cout_real + Z_Lout_real));
%% Semiconductor Losses
%FET & Diode Characteristics
t_r = 25*10^(-9);
t_f = 10*10^(-9);
Rdson = 0.016 * 2.2; %take into account x2.2 power dissipation
V_diode = 0.5;
% FET and Diode Losses
P_cond_FET = max(I_out)^2 * max(Dccm) * Rdson; % Conduction Loss:
P_sw_FET = 1/2*(t_f+t_r) * max(I_out) * max(Vin)*fsw; % switching loss
P_sw_total = P_cond_FET + P_sw_FET;
P_diode = max(V_diode*I_out.*(1-Dccm));
%% Thermal Evaluation
% Input Inductor
T_Lin = (P_AC_core_in + P_core_in + P_DC_core_in) * 23 + 50;
% Output Inductor
T_Lout = (P_AC_core_out + P_core_out + P_DC_core_out) * 23 + 50;
% Use ONE heatsink for both devices. Heatsink: 48.6 C / 6W. = 8.1 C/W
% Diode
R_Diode_JA = 32.31;
T_diode = 40 + R_Diode_JA*P_diode %

```

```

% FET
R_FET_JA = 27.254;
T_FET = 40 + R_FET_JA*(P_cond_FET + P_sw_FET)
fprintf("")
%% Evaluation Performance
fprintf("Performance Evaluation \n")
fprintf("Condition: \n")
fprintf("Power [W]: " + PWR + "\n")
fprintf("Input Voltage [V]: " + Vin + "\n")
fprintf("Switching Frequency [Hz]: " + fsw + "\n")
fprintf("\n \n")
fprintf("Input Filter and EMI Consideration" + "\n")
fprintf("Maximum Allowed Filter Resonant Frequency (the design should be smaller) [rad/s]: " + win_target + "\n")
fprintf("Filter Resonant Frequency [rad/s]: " + (1/sqrt(Lin*Cin)) + "\n")
fprintf("Characteristics Filter Impedance [Ohm]: " + (sqrt(Lin/Cin)) + "\n")
fprintf("Equivalent Negative Converter Load [Ohm]: " + R_conv_neg + "\n")
fprintf("Maximum Allowed FR Attenuation Factor [dB] @ switching frequency: " + EMI_TF_dB_attenuation_factor + "\n")
fprintf("See Attached Filter FR for attenuation evaluation @ switching frequency" + "\n")
fprintf("\n \n")
fprintf("Ripple Evaluations: \n")
fprintf("Output Voltage Ripple [Vpp], Limit 80mVpp: " + v_Co_rip_pp + "\n")
fprintf("Input Current Ripple [A_pp], Limit 50mApp: " + i_input_rippl + "\n")
fprintf("\n \n")
fprintf("Inductor Specification: \n")
fprintf("Input Inductor Value [H]: " + Lin + "\n")
fprintf("Input Inductor Magnetix Flux Density, limit 0.3T [T]: " + B_core_in + "\n")
fprintf("Input Inductor Current Density, limit 5000000 A/m^2 [A/m^2]: " + Lin_current_density + "\n")
fprintf("Output Inductor Value [H]: " + Lout + "\n")
fprintf("Input Inductor Magnetix Flux Density, limit 0.3T [T]: " + B_core_out + "\n")
fprintf("Output Inductor Current Density, limit 5000000 A/m^2 [A/m^2]: " + Lout_current_density + "\n")
fprintf("\n \n")
fprintf("Power Loss and Efficiency Evaluation: \n")
fprintf("FET Conduction Loss [W]: " + P_cond_FET + "\n")
fprintf("FET Switching Loss [W]: " + P_sw_FET + "\n")
fprintf("Diode Loss [W]: " + P_diode + "\n")
fprintf("Input Filter Inductor Core Loss [W]: " + P_core_in + "\n")
fprintf("Input Filter Inductor Conduction Loss DC [W]: " + P_DC_core_in + "\n")
fprintf("Input Filter Inductor Conduction Loss AC w/ Skin Depth [W]: " + P_AC_core_in + "\n")
fprintf("Output Filter Inductor Core Loss [W]: " + P_core_out + "\n")
fprintf("Output Filter Inductor Conduction Loss DC [W]: " + P_DC_core_out + "\n")
fprintf("Output Filter Inductor Conduction Loss w/ Skin Depth [W]: " + P_AC_core_out + "\n")
fprintf("Total Loss [W]: " + (P_cond_FET + P_sw_FET + P_diode + P_core_in + P_AC_core_in + P_core_out + P_AC_core_out) + "\n")
fprintf("Given Loss Budget [W]: " + PWR * 0.1 + "\n")
fprintf("Efficiency, Requirement > 0.9: " + ((PWR-(P_cond_FET + P_sw_FET + P_diode + P_core_in + P_AC_core_in + P_core_out + P_AC_core_out + P_DC_core_in + P_DC_core_out))/PWR) + "\n")
fprintf("\n \n")
fprintf("Thermal Performance Evaluation: \n")
fprintf("Designed Effective FET Junction To Ambient Designed Thermal Impedance [C/W]: " + R_FET_JA + "\n")
fprintf("FET Calculated Junction Temperature, Limit @ 150C [C]: " + T_FET + "\n")
fprintf("Designed Effective Diode Junction To Ambient Designed Thermal Impedance [C/W]: " + R_Diode_JA + "\n")
fprintf("Diode Calculated Junction Temperature, Limit @ 150C [C]: " + T_diode + "\n")
fprintf("Input Inductor Temperature (Assuming 50C Ambient) [C]: " + T_Lin + "\n")
fprintf("Output Inductor Temperature (Assuming 50C Ambient)[C]: " + T_Lout + "\n")
fprintf("\n \n")
fprintf("Transient and Steady State Tolerance Performance: \n")
fprintf("See Controller Section of the Report \n")
fprintf("\n \n")

```

Appendix II: Feedback Controller Code

```
clear;
close all;
clc;
%%
P_arr = [50, 150];
Vin_arr = [22,32];
D_arr = [15/22, 15/32];
R_arr = [15^2/50, 15^2/150]; %  $V^2 / R = P \rightarrow R = V^2/P$ 
%% Value Select
s = tf('s');
%sequence 32 50 32 150 22 50 22 150
Vin = 32;
P = 50;
R = 15^2/P;
D = 15/Vin;
L = 25.5 * 10^-6;
Co = 39 * 10^-6;
Resr_Co = 0.025;
Z_Co = 1/(Co*s) + Resr_Co;
%% Transfer Function Graphing
s = tf('s');
LC_transfer = (1/Z_Co + 1/R)^(-1) / ((1/Z_Co + 1/R)^(-1) + L*s);
TF_d_vo = Vin / (L*Co*s^2 + L/R*s + 1);
TF_vi_vo = D / (L*Co*s^2 + L/R*s + 1);
%% Controller Design
s = tf('s');
% Kp
K_p = 2200/5200; %setting the crossover
%Integral_
w_lag = (4.7*10^-6)*680;
Integral = 1/(w_lag*s);
%LPF 1
LPF1 = (10000/10000) * 1/(10000*6.8*10^-9*s + 1);
%LPF2 - identical to LPF 1
LPF2 = (10000/10000) * 1/(10000*6.8*10^-9*s + 1);
% Controller Creation
CTRL = K_p * Integral * LPF1 * LPF2 ;
% figure()
% bode(CTRL);
%% Margin Calculation
FW_d_vo = CTRL * TF_d_vo;
%FW_vi_vo = CTRL * TF_vi_vo;
figure('Name', "Case: " + Vin + "[V], " + P + "[W]")
title("Case: " + Vin + "[V], " + P + "[W]")
margin(FW_d_vo)
% hold on
% bodeTF_d_vo)
figure(2)
bode(CTRL)
legend("D CTRL", "CTRL")
close_loop = FW_d_vo/(1+FW_d_vo);
figure(3)
bode(close_loop)
figure('Name', "Case: " + Vin + "[V], " + P + "[W]")
hold on;
step(close_loop)
figure()
bode(TF_d_vo)
```